INFOMDW: Data Wrangling

Heterogeneous Data Integration

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Data integration

- Organizations use databases: mainframes, workstations, servers
  - Built from scratch each time… usually
  - Many models of the same object
- Need for information sharing across databases
  - Exploit advances in distributed computing and networking

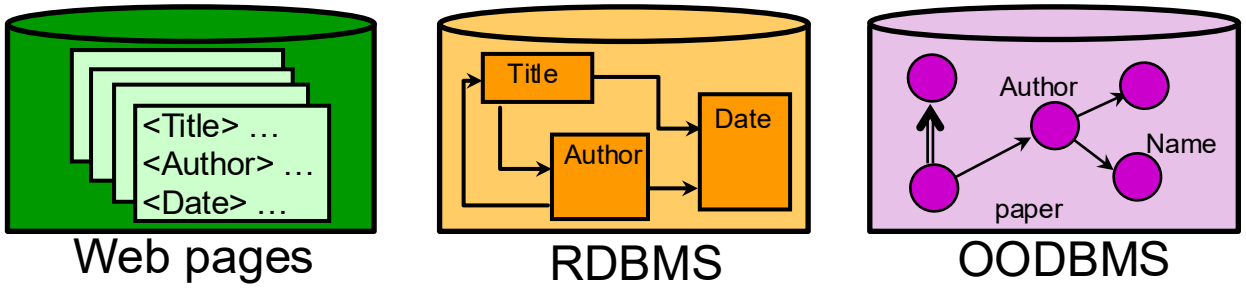
Data integration

- Fact: Databases are managed by different persons
- Challenge 1: Provide uniform access to the users
- Challenge 2: Allow DB to interoperate but …
  - Autonomous
  - Different OS
  - Different purposes
  - Different data modeling
  - Different data formats
  - Different access and communication protocols

Information is everywhere

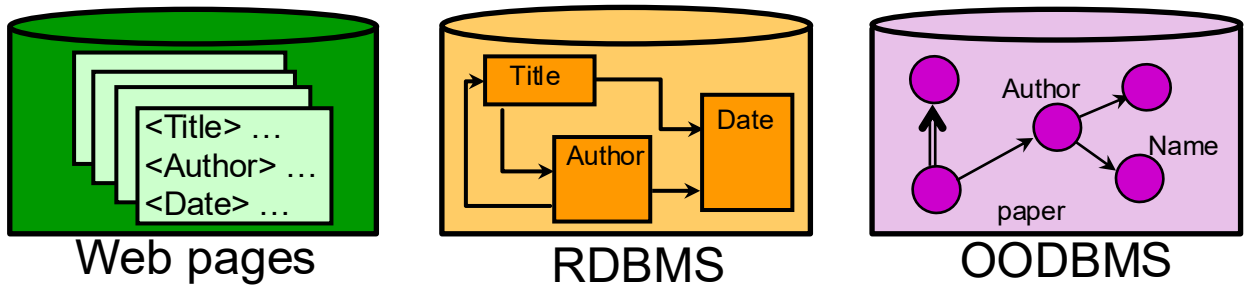


- Databases .. but not only
- Mail messages
- Web pages
- Spreadsheets
- Text documents
- Multimedia annotations
- XML Documents



There are many needs

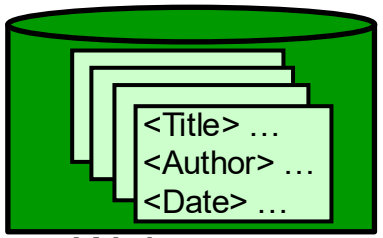
- Exchanged
- Replicated
- Migrated
- Integrated
- Adapted



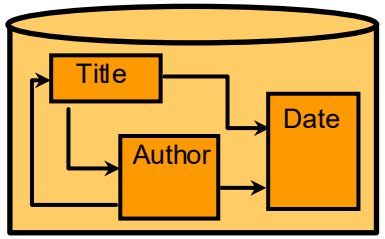
Data in different forms



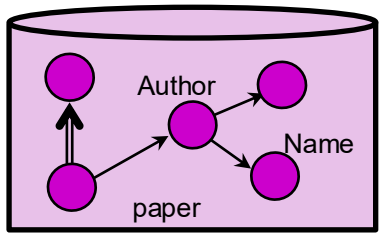
- Personalization: content adapted to user
  - upon system's decision
  - upon user's request
- Customization: structure adapted to user
  - according to the user's role
  - upon user's request
- System Dependent
  - Performance Reasons
- Context dependence
  - User, Device, Network, Place, Time, Rate



Web pages



RDBMS

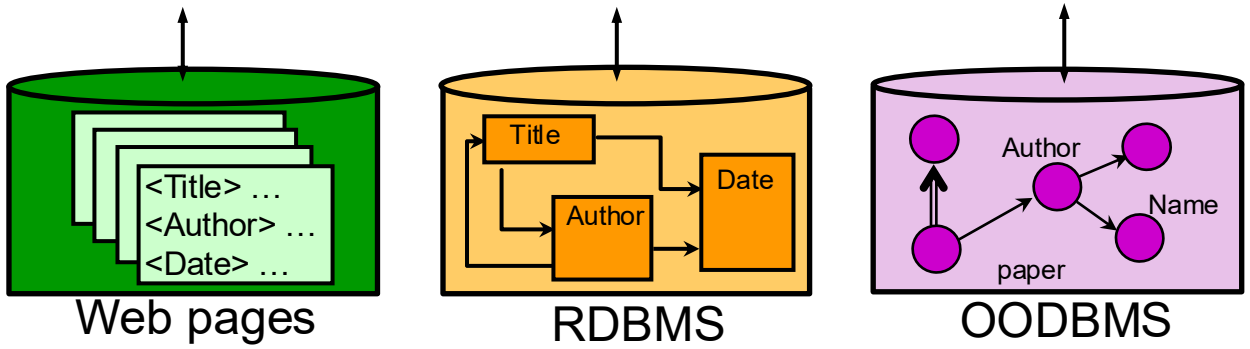


OODBMS

System level



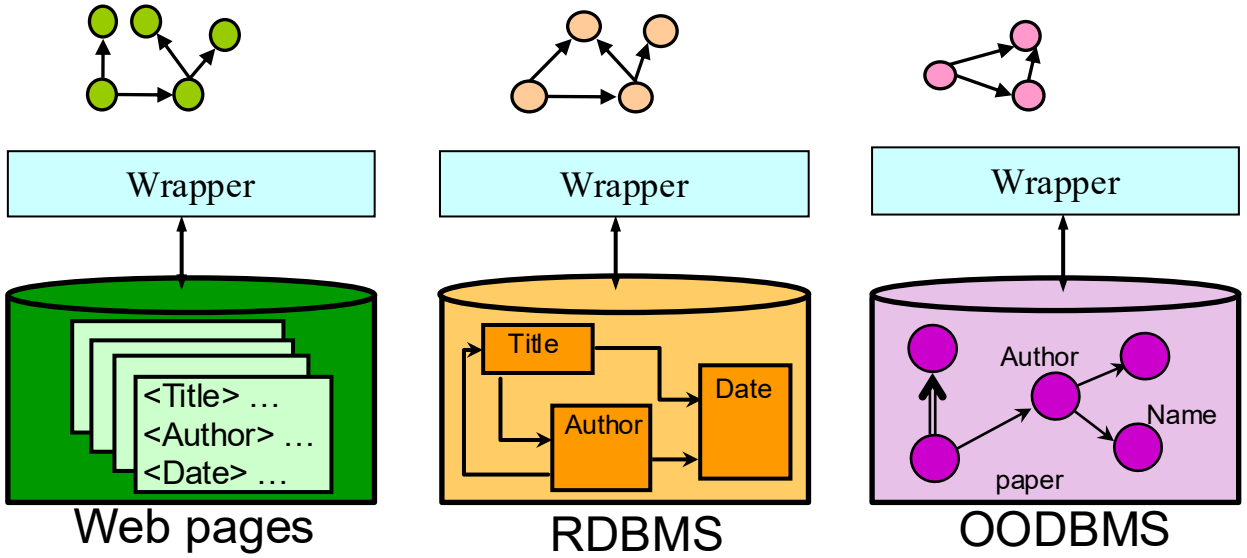
- Protocols
  - Predefined methods of communication



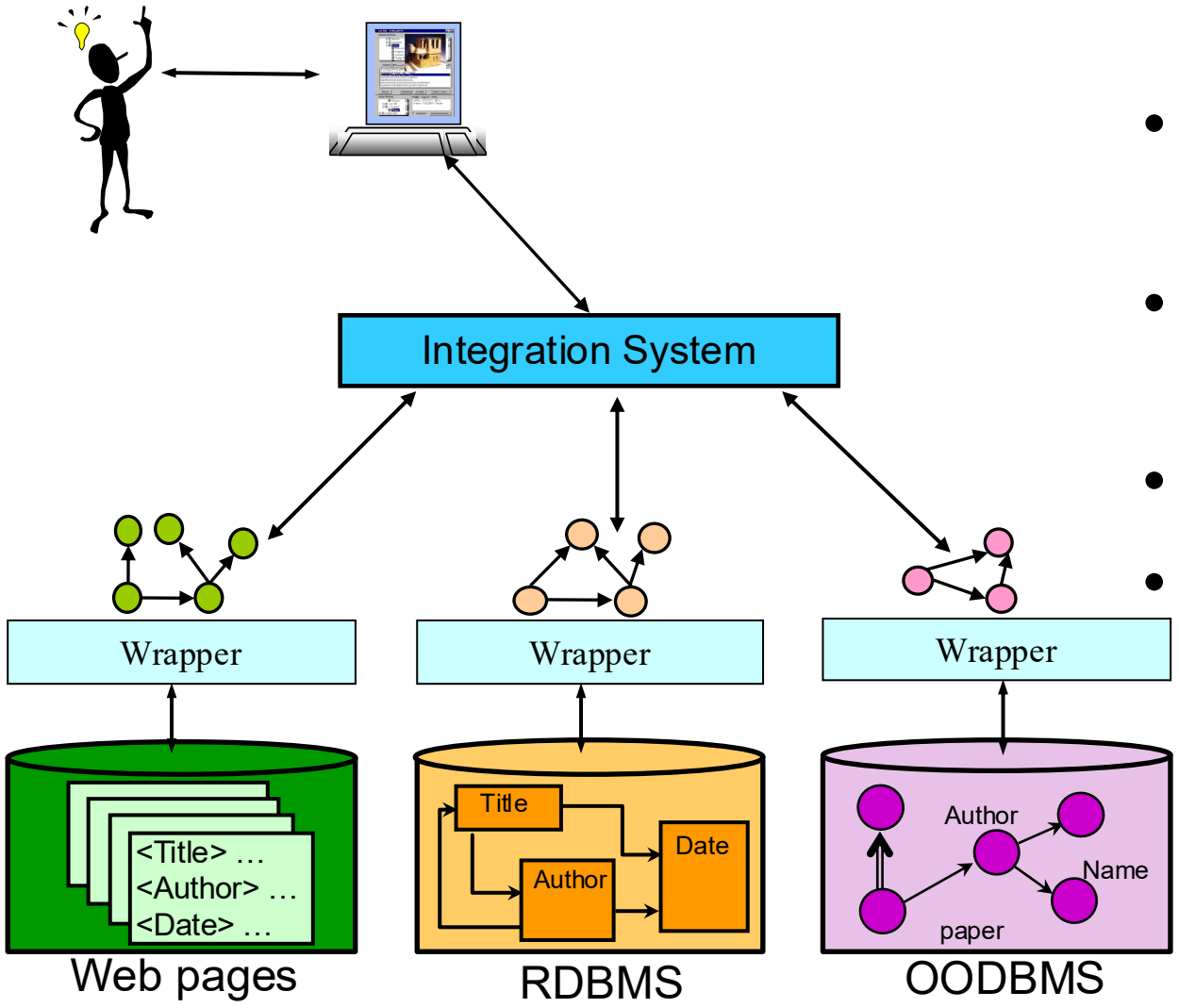
Model level



- Wrappers
  - Wrap the sources into a common data model



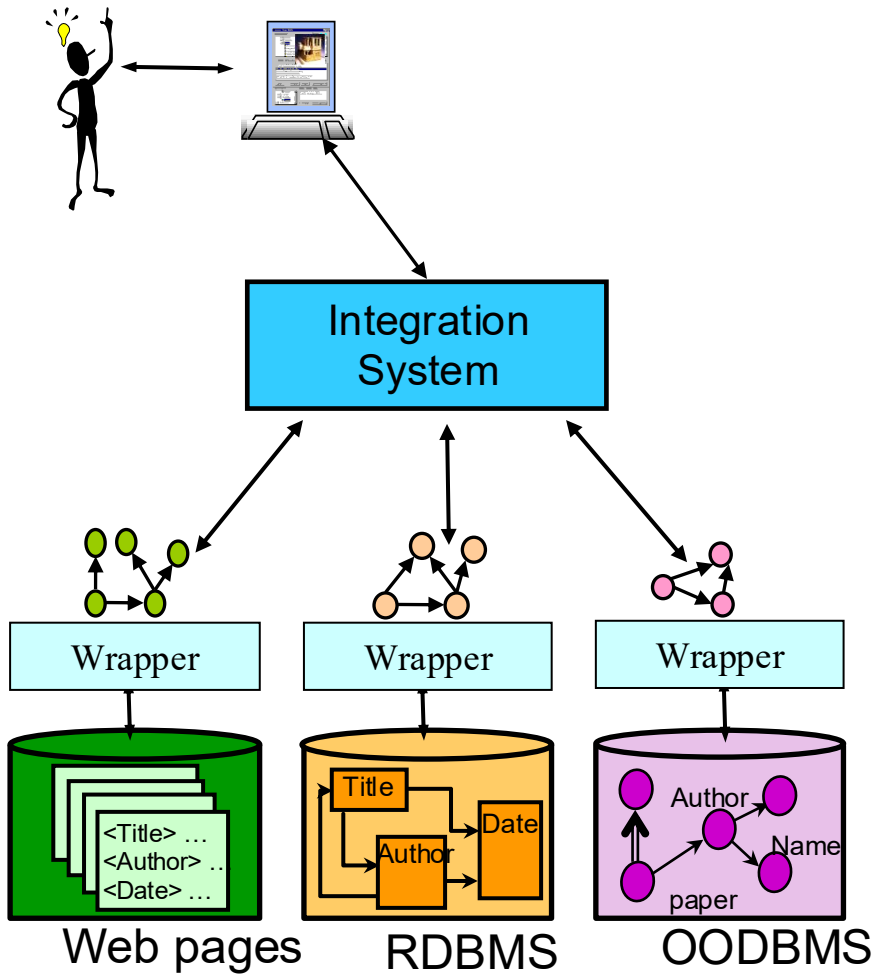
Structural and semantic level



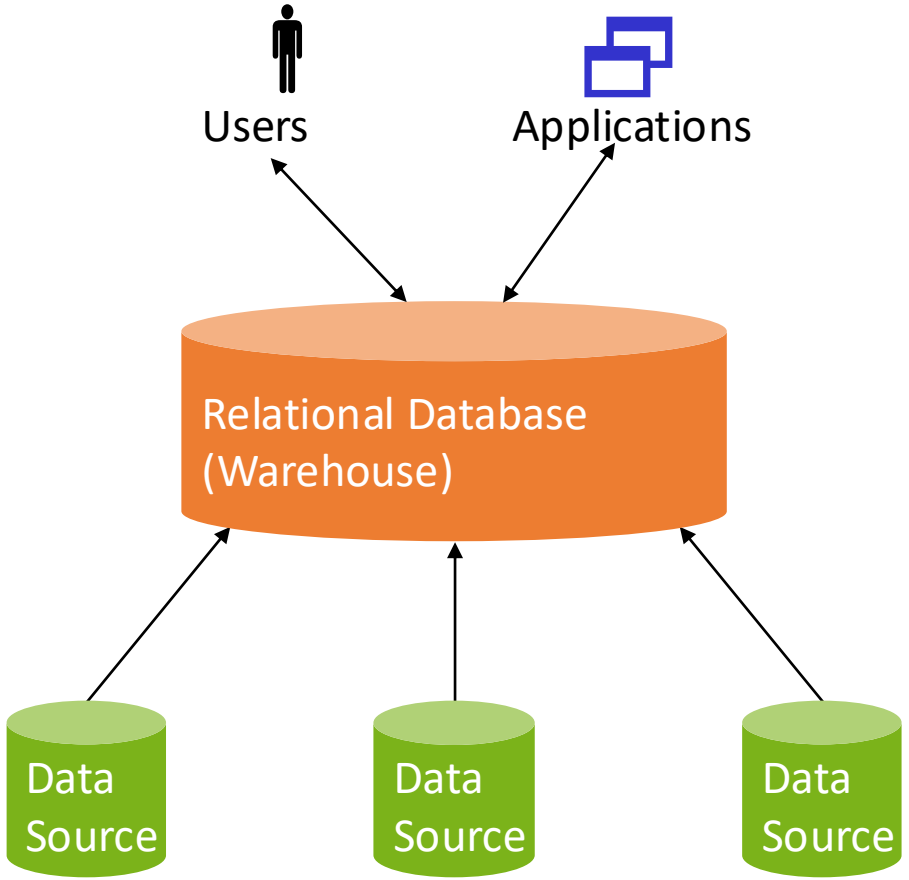
- Structural
  - Different structures for the same data
- Semantic
  - Different semantics of the same structure
- Integrated View
- Many architectures for this basic idea

An information integration system

- A set of Local Databases
  - Each with a Local Schema & a Local Instance
- A Global Integrated Schema
- A set of Mappings
  - Between the Global and the Local Schemas
  - Describe the relationship between the data in the sources and the data as the user “sees” them



Data warehouse architecture



OLAP / Decision Support
Data Cubes / Data Mining

ETL Tools
(Extract-Transform-Load)

Data Cleaning

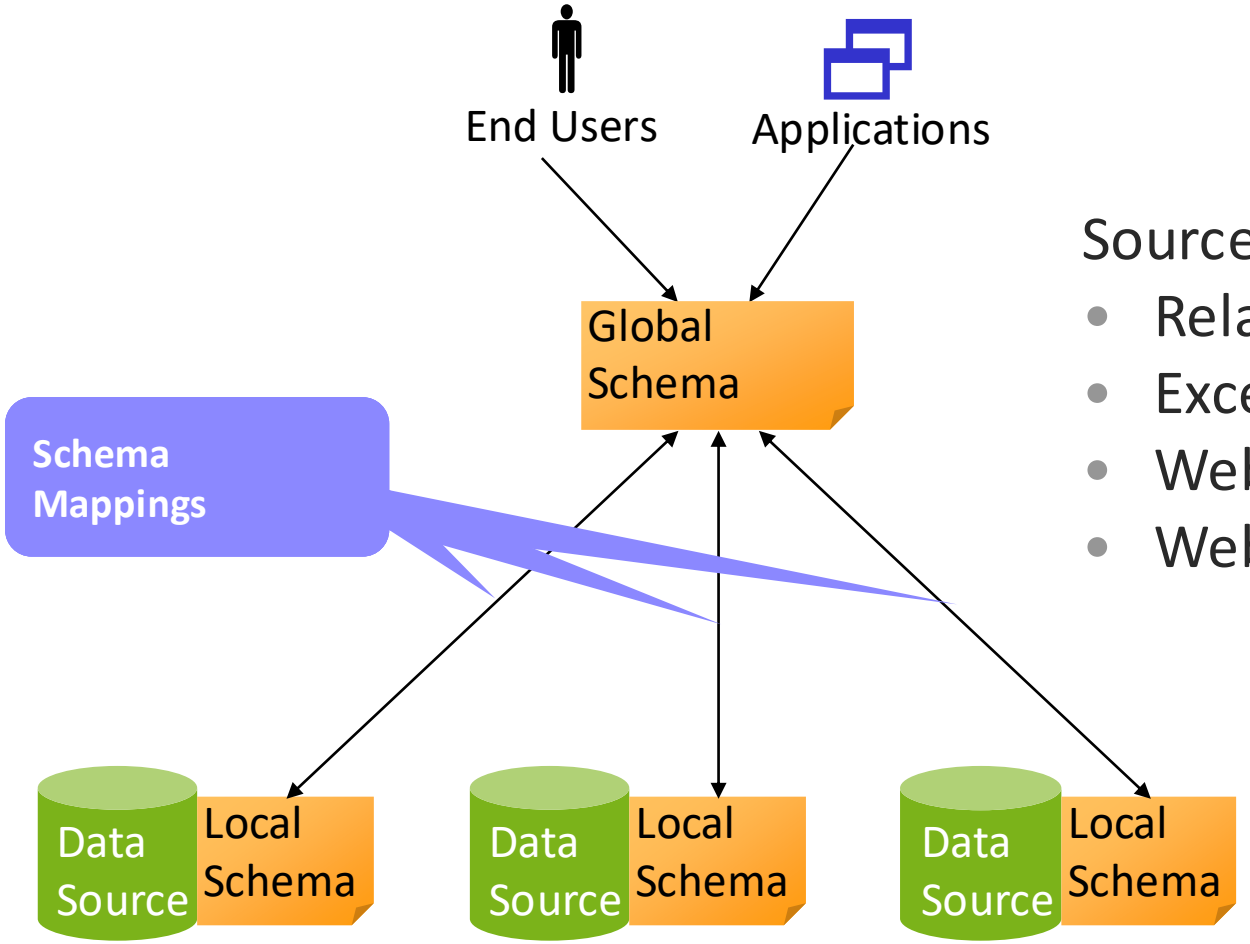
Virtual integration architecture

- Leave the data in the sources
- When a query comes in:
  - Determine the relevant sources to the query
  - Break down the query into sub-queries for the sources
  - Get the answers from the sources, filter them if needed and combine them appropriately
- Data is fresh
- Otherwise known as

On Demand Integration

Virtual integration architecture

Design-Time

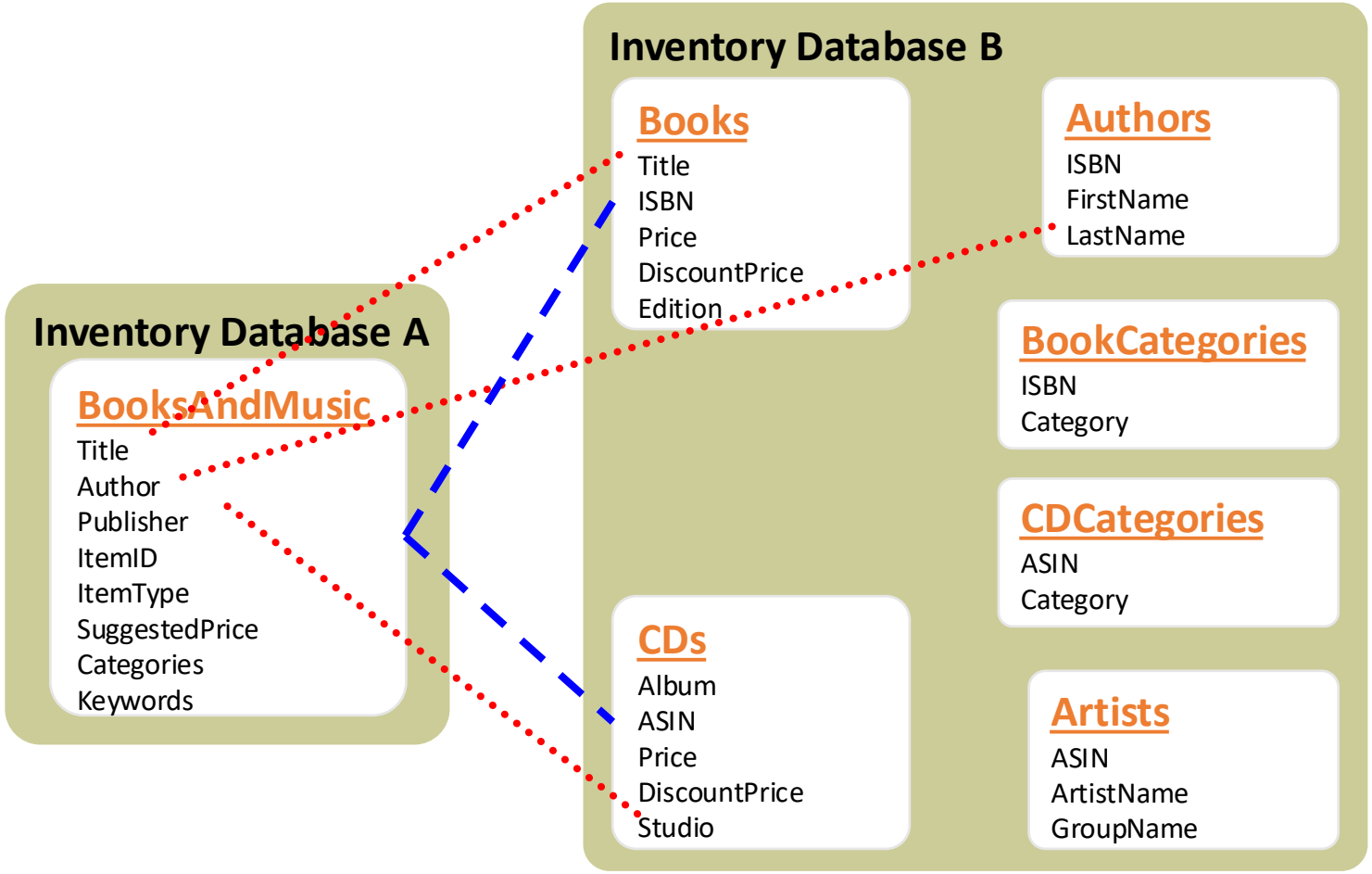


Sources can be:

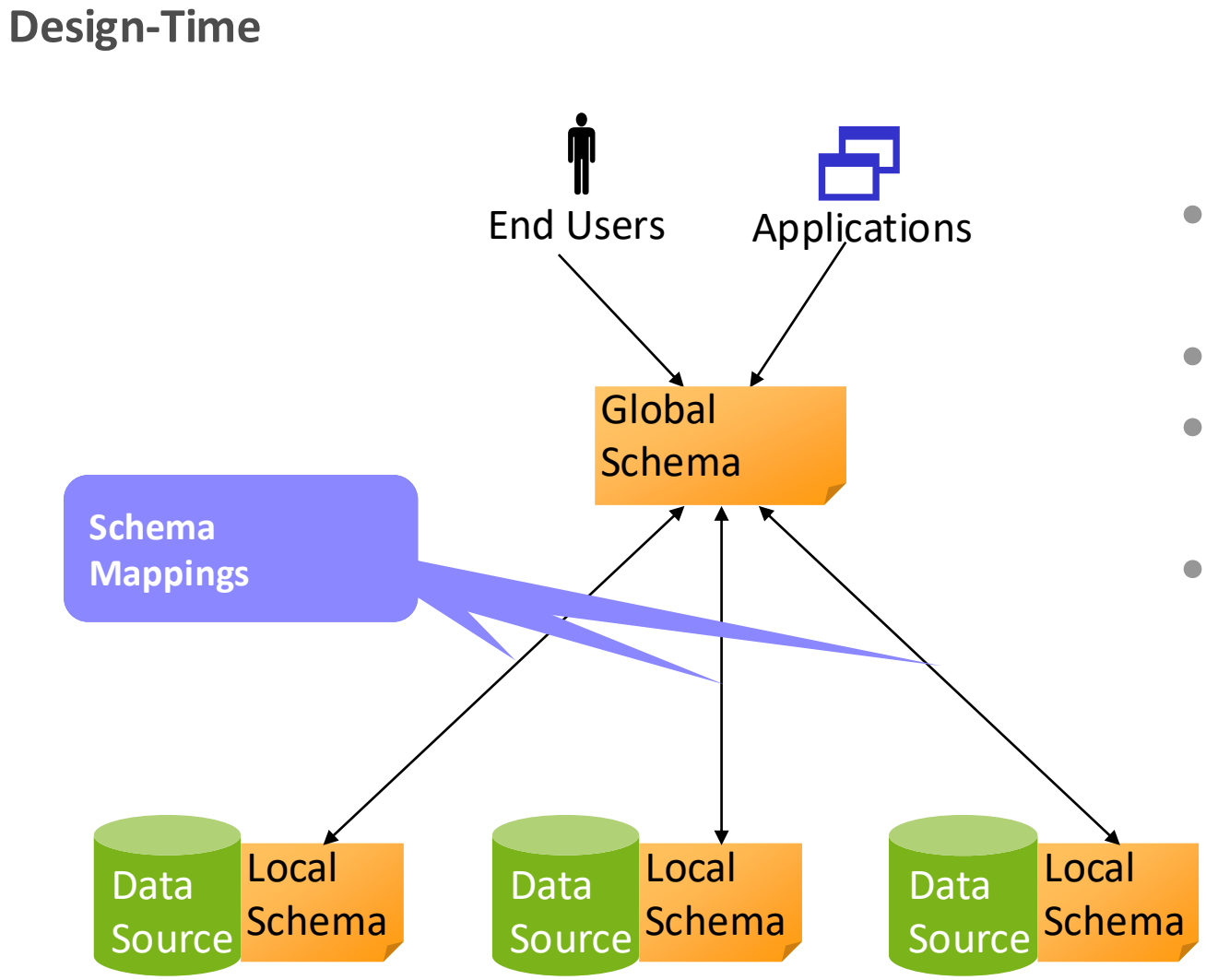
- Relational DBs
- Excel Files
- Web Sites
- Web Services

Schema mappings

- Differences in:
  - Names in schema
  - Attribute grouping
  - Coverage of databases
  - Granularity and format of attributes



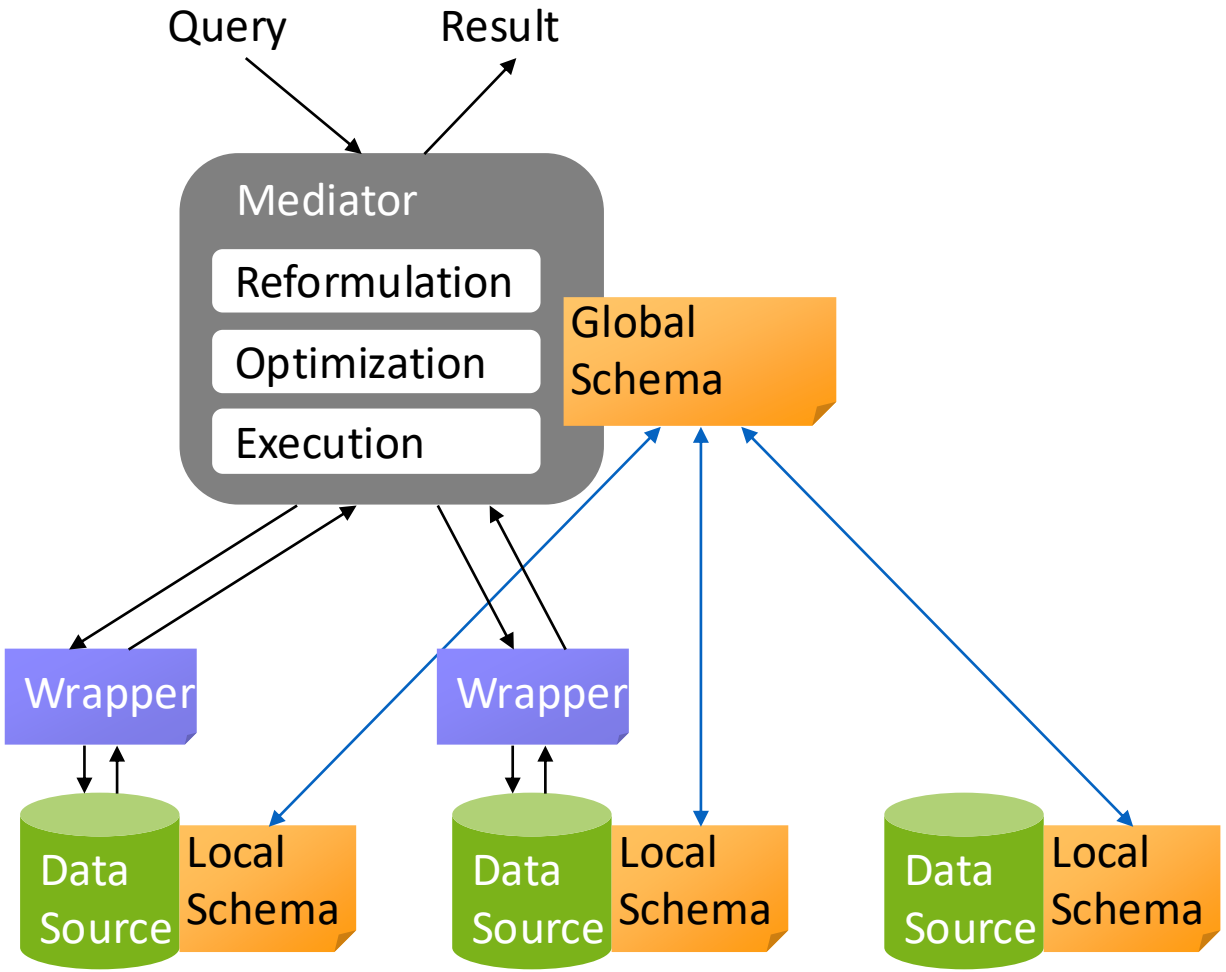
Issues for schema mappings



- What formalisms to express them?
- How to create them?
- Can we discover them somehow?
- How do we use them?

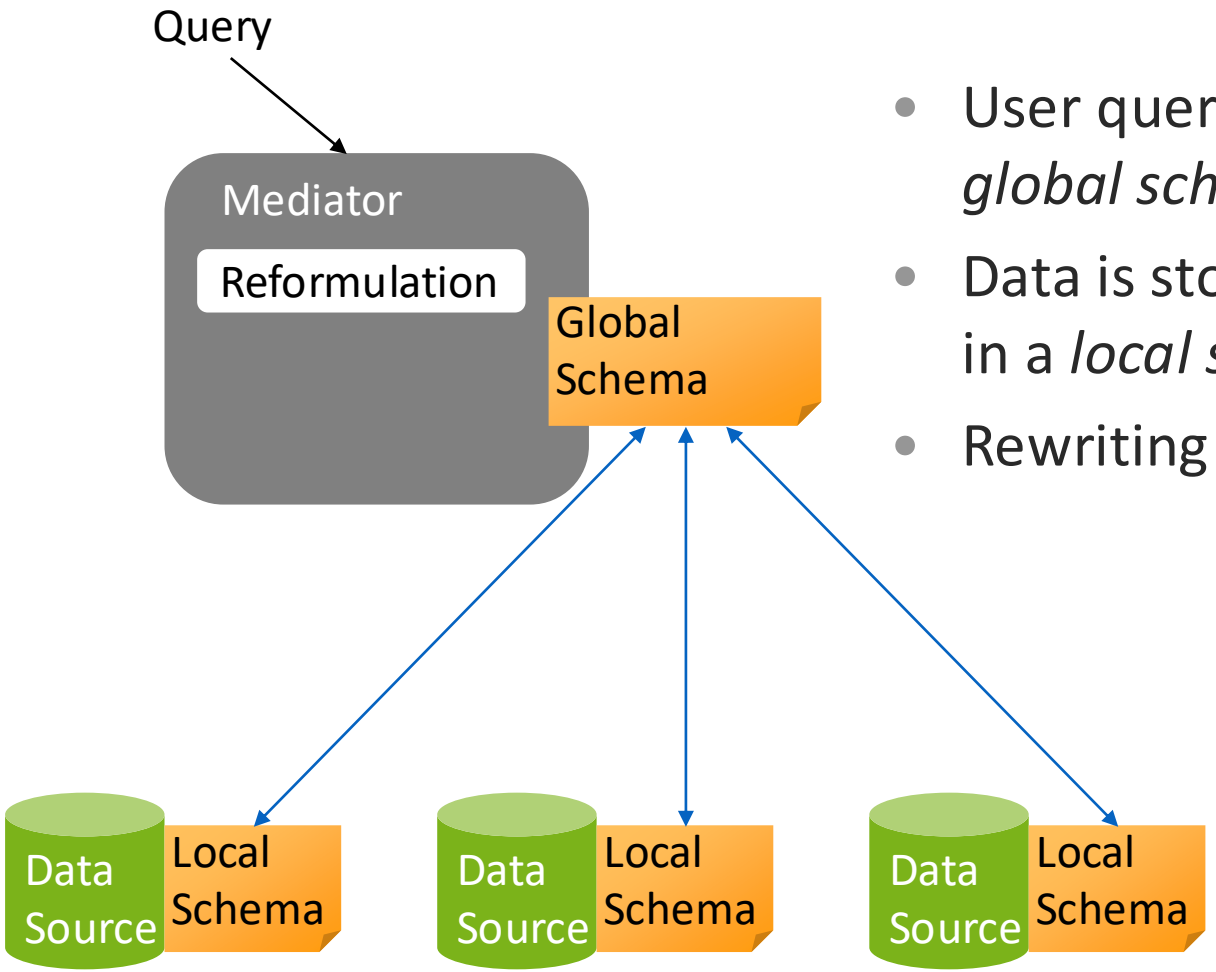
Virtual integration architecture

Run-Time



Issues for query processing

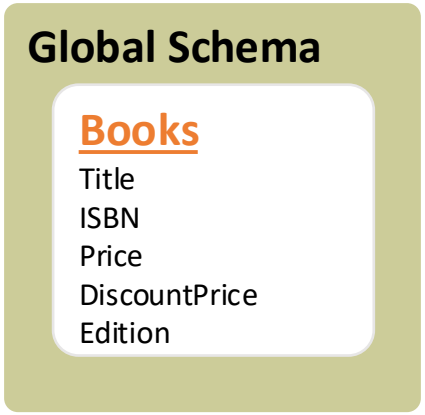
Reformulation

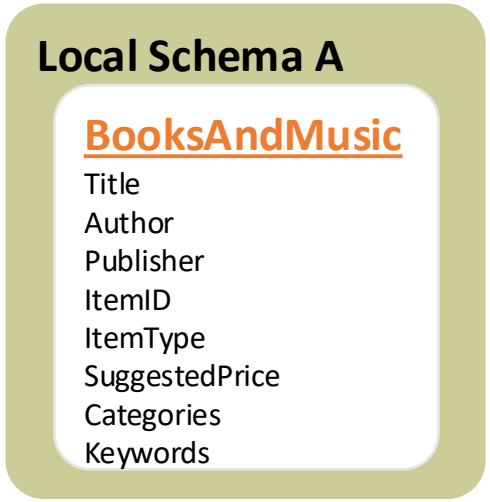


- User queries refer to the *global schema*
- Data is stored in the sources in a *local schema*
- Rewriting algorithms

Issues for query processing

Reformulation





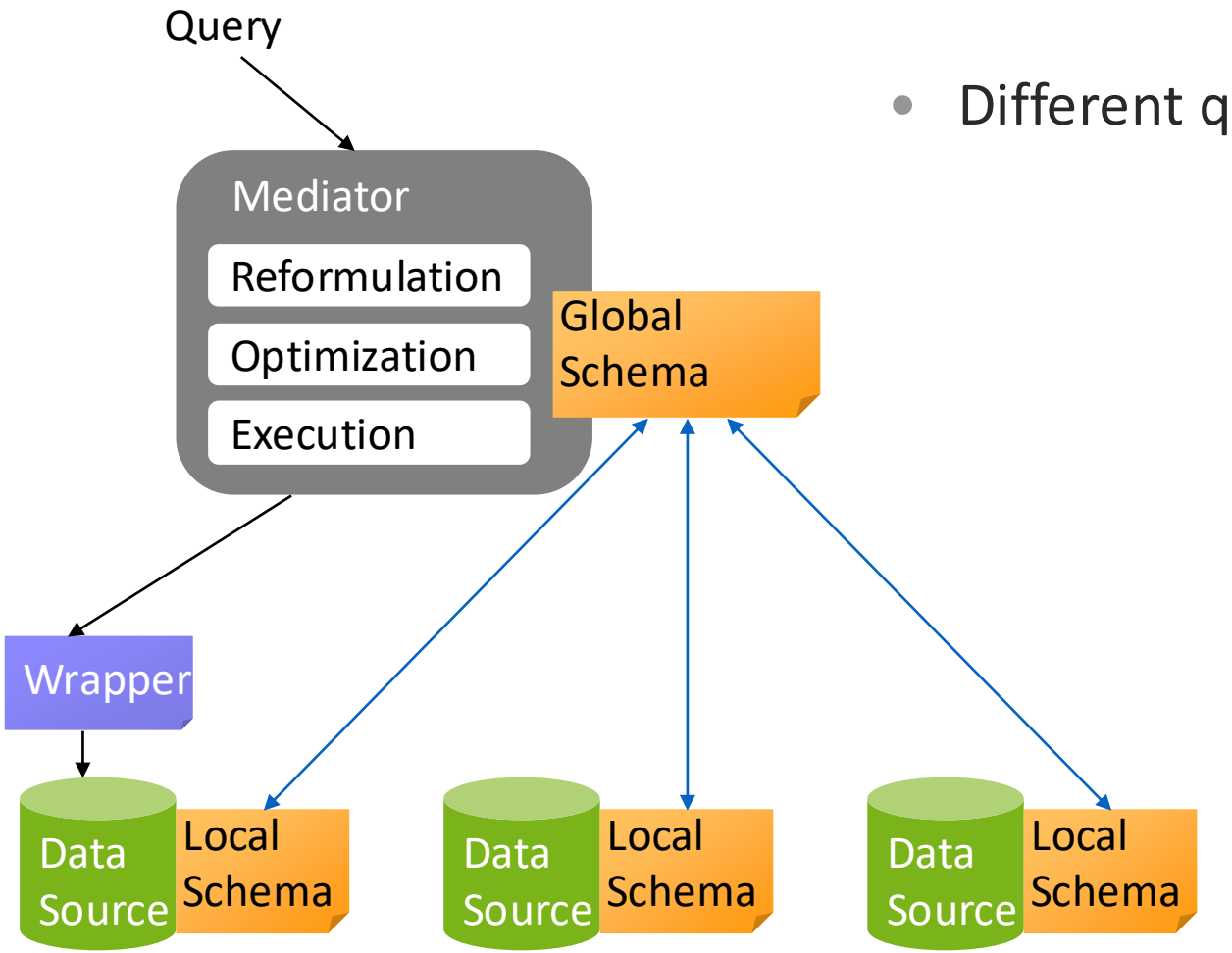

```
SELECT ISBN, Price
FROM Books
WHERE Title = 'on the road'
```



```
SELECT ItemID, SuggestedPrice
FROM BooksAndMusic
WHERE Title = 'on the road'
AND ItemType = 'Books'
```

Issues for query processing

Query Translation



- Different query languages

Issues for query processing

Query Translation

Global Schema

Books
Title
ISBN
Price
DiscountPrice
Edition

```
SELECT ISBN, Price
FROM Books
WHERE Title = 'on the road'
```

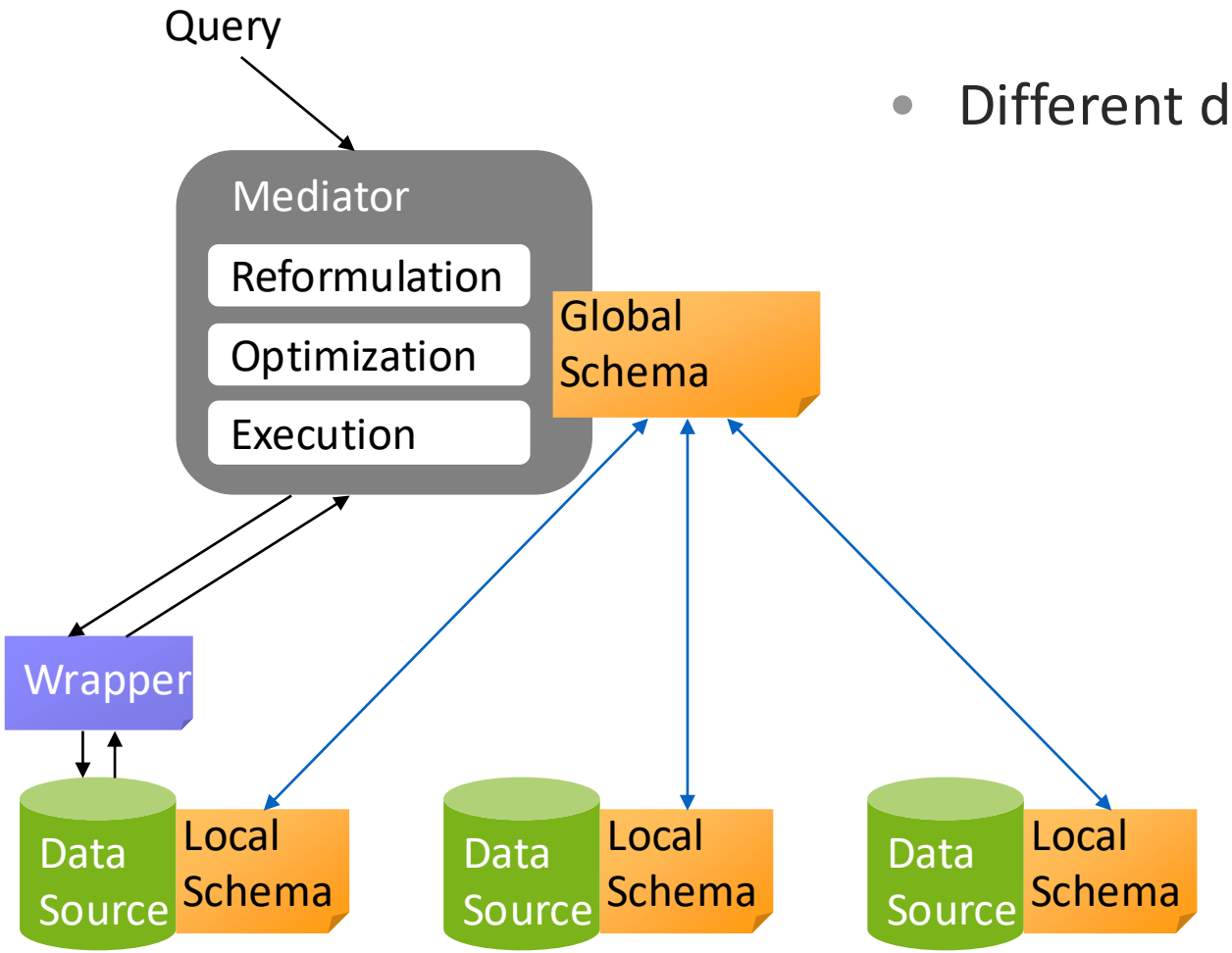
Local Source A



http://www.amazon.com/homepage.html?ItemType=Books&Title=on+the+road

Issues for query processing

Data Translation



- Different data models

Issues for query processing

Data Translation

Global Schema

Books
Title
ISBN
Price
DiscountPrice
Edition

Local Result A



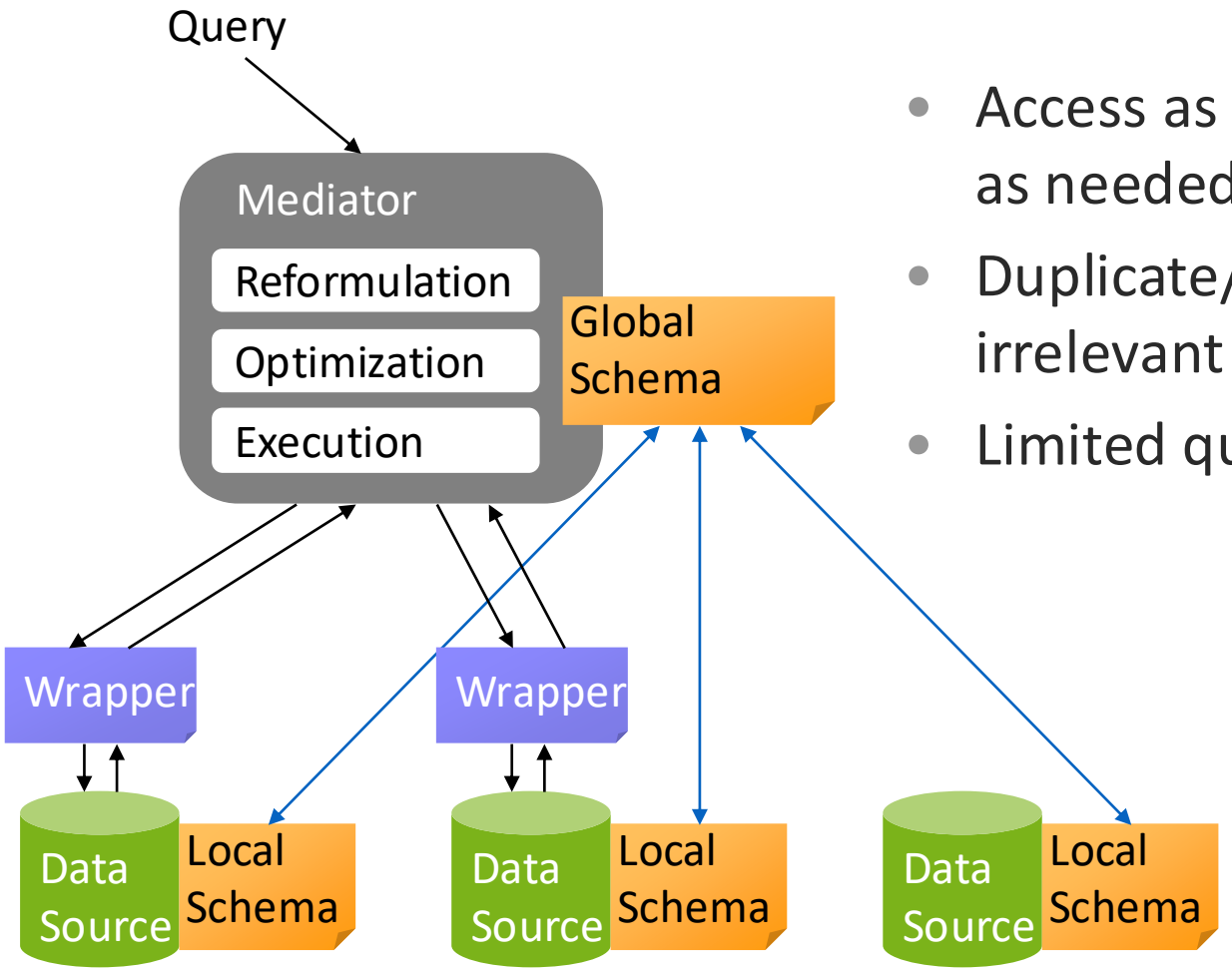
On the Road -- by Jack Kerouac; Paperback (Rate it)
Buy new: $10.86 -- Used & new from: $5.90

Title	ISBN	Price	...	...
On the Road	123	10.86	...	...

```
<table>
  <tr>
    <td>
      <a href=/details?isbn=123>
        <b>On the Road</b>
      </a>
      -- by Jack Kerouac; Paperback
      <br>
      <a href=/details?isbn=123>
        Buy new
      </a>
      :<b class=price>$10.86</b>
    </td>
  </tr>
</table>
```

Issues for query processing

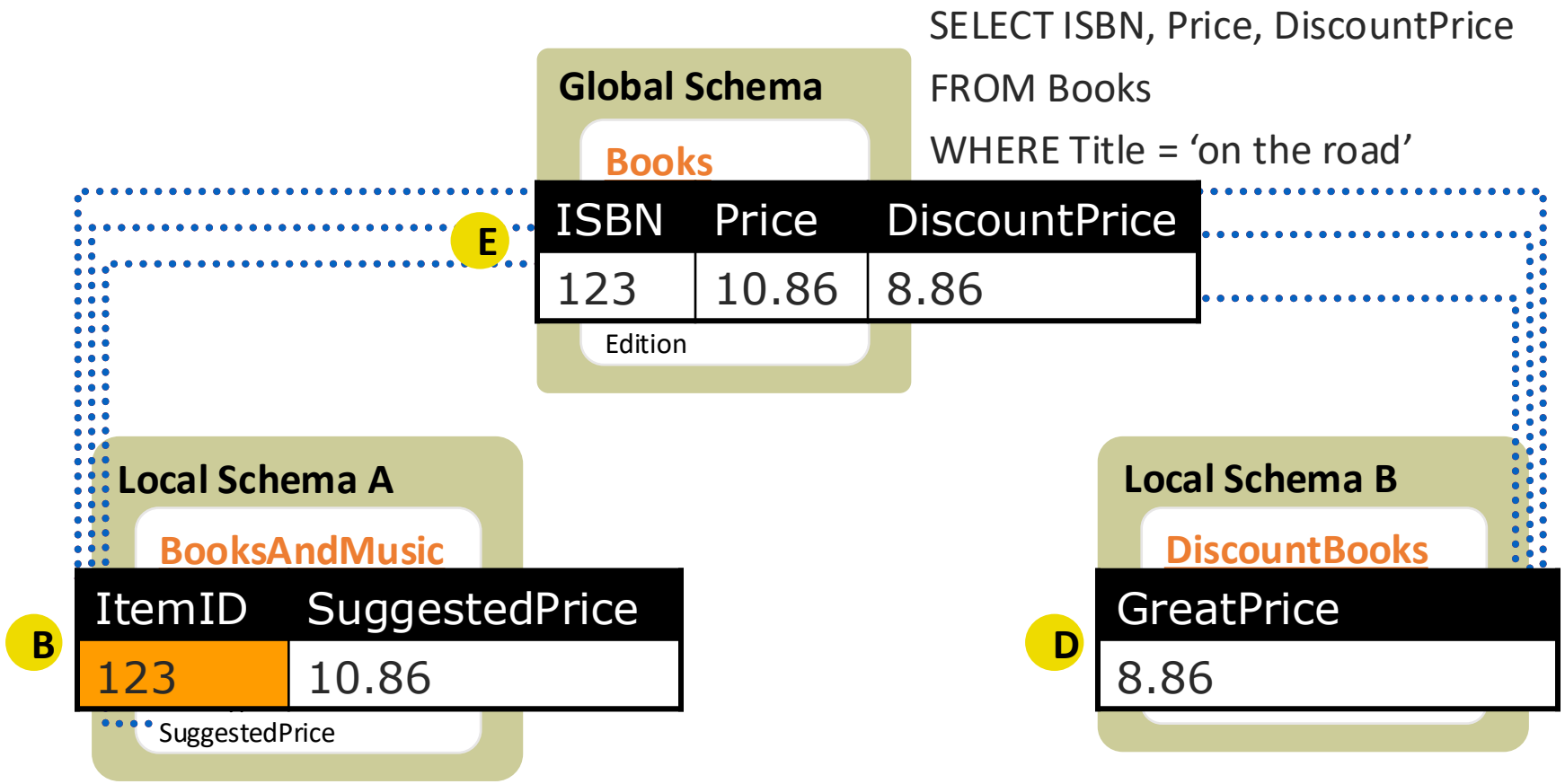
Query Execution



- Access as many data sources as needed
- Duplicate/redundant and irrelevant data
- Limited query capabilities

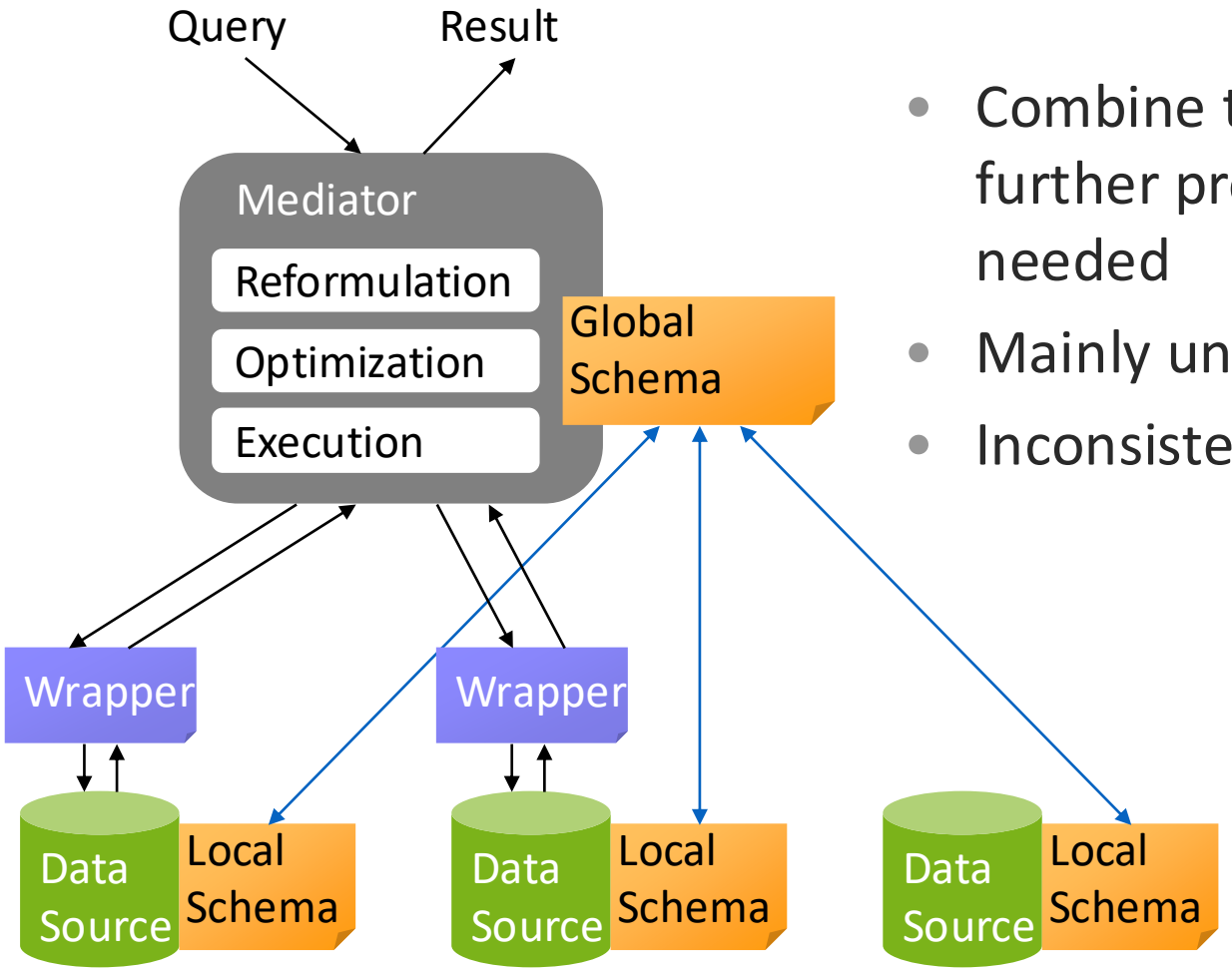
Issues for query processing

Limited Query Capabilities



Issues for query processing

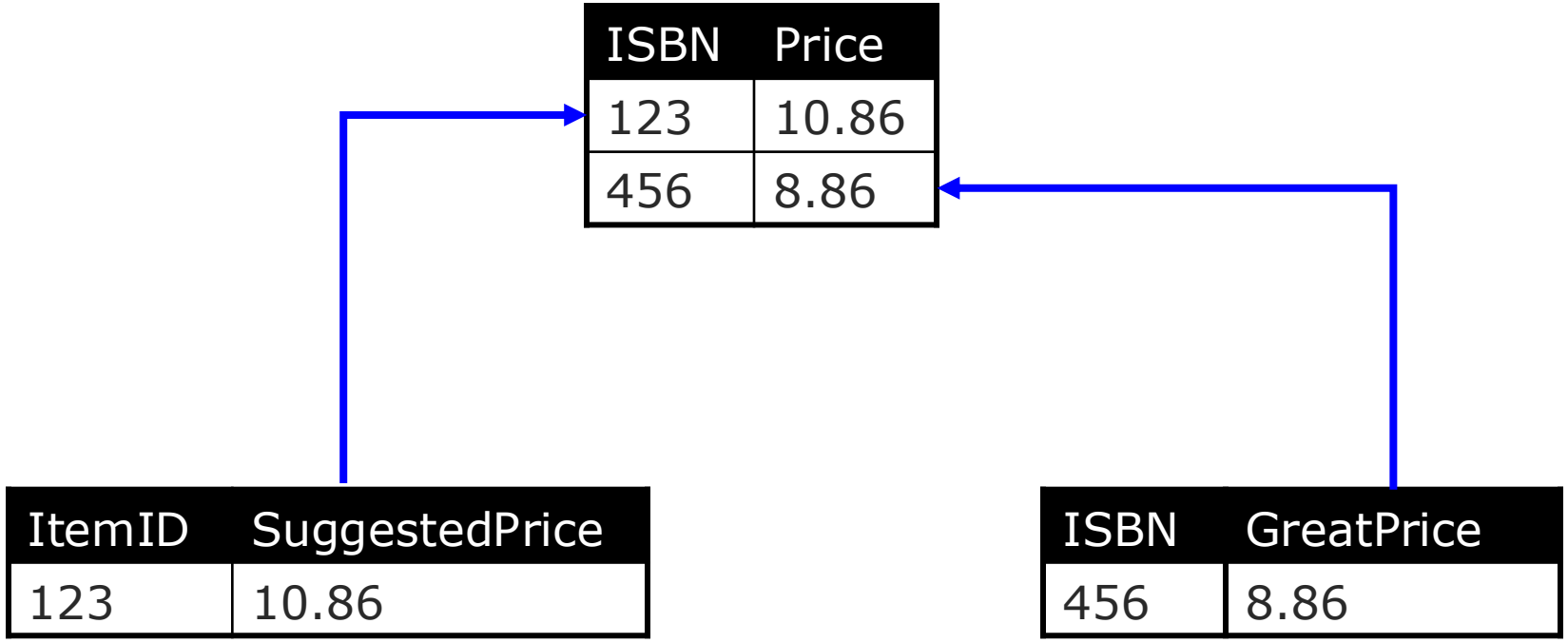
Query Answering



- Combine the results and further process them if needed
- Mainly union and merge
- Inconsistencies

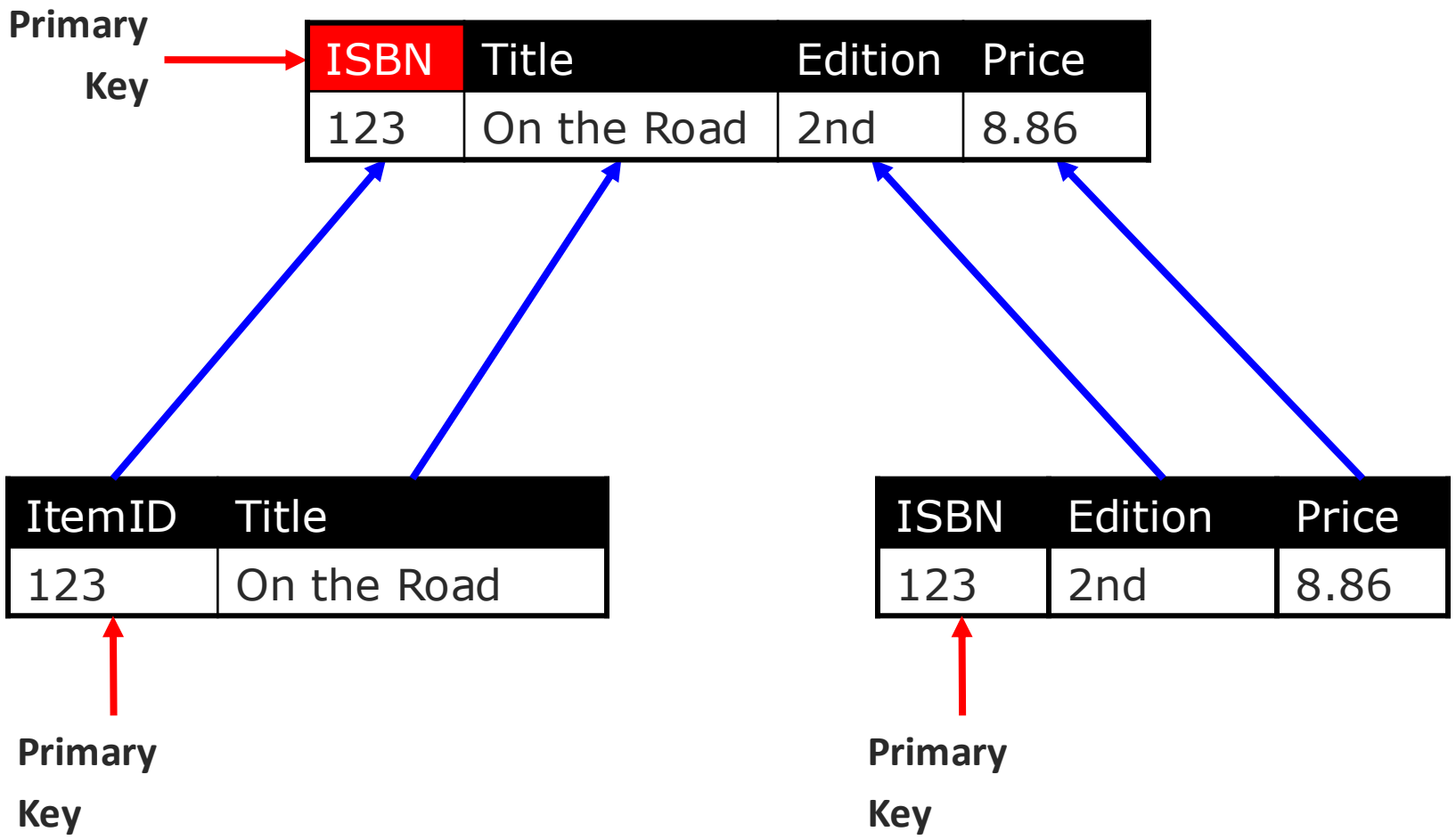
Issues for query processing

Query Answering (Union)



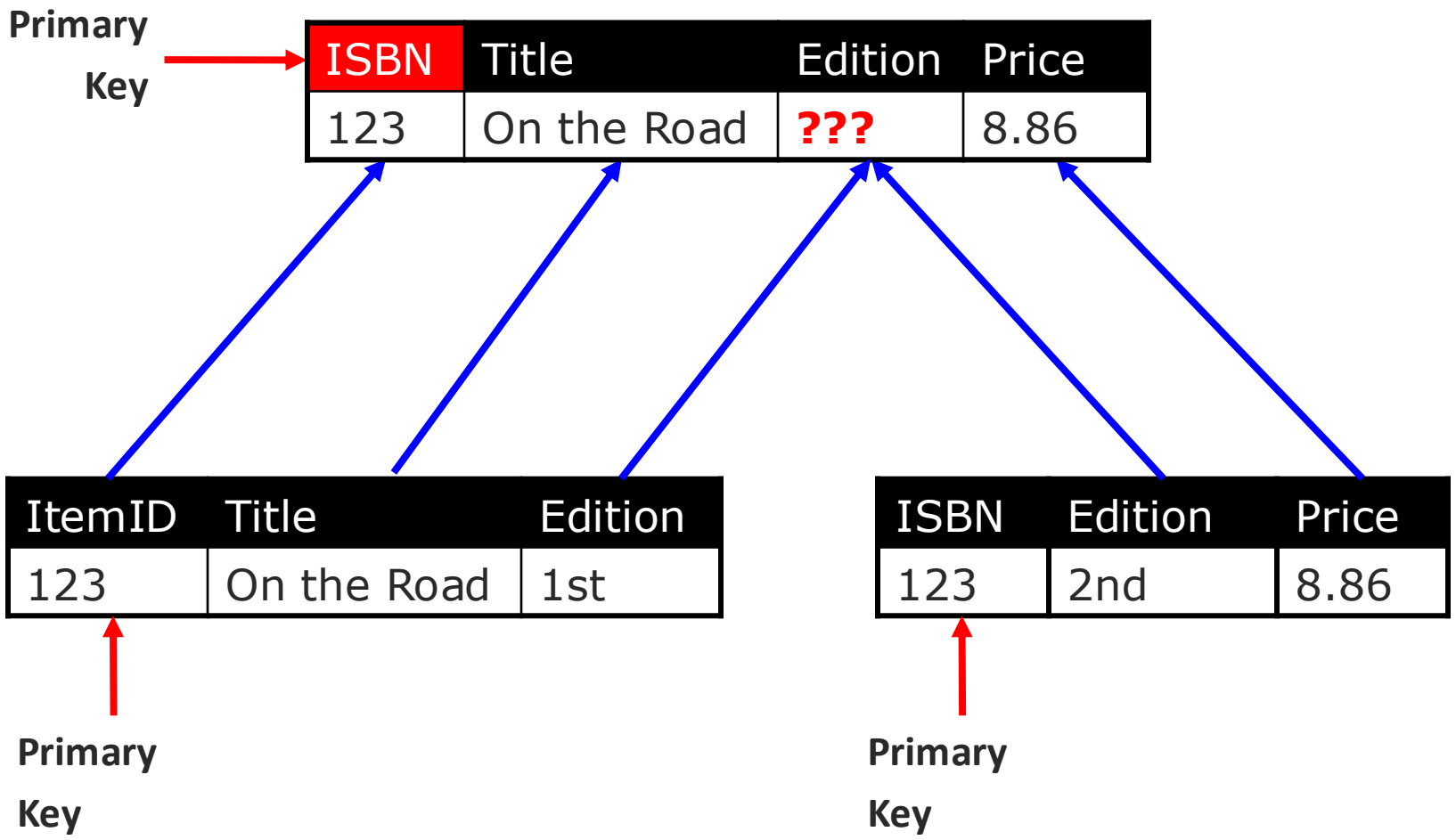
Issues for query processing

Query Answering (Merge)



Issues for query processing

Query Answering (Inconsistencies)



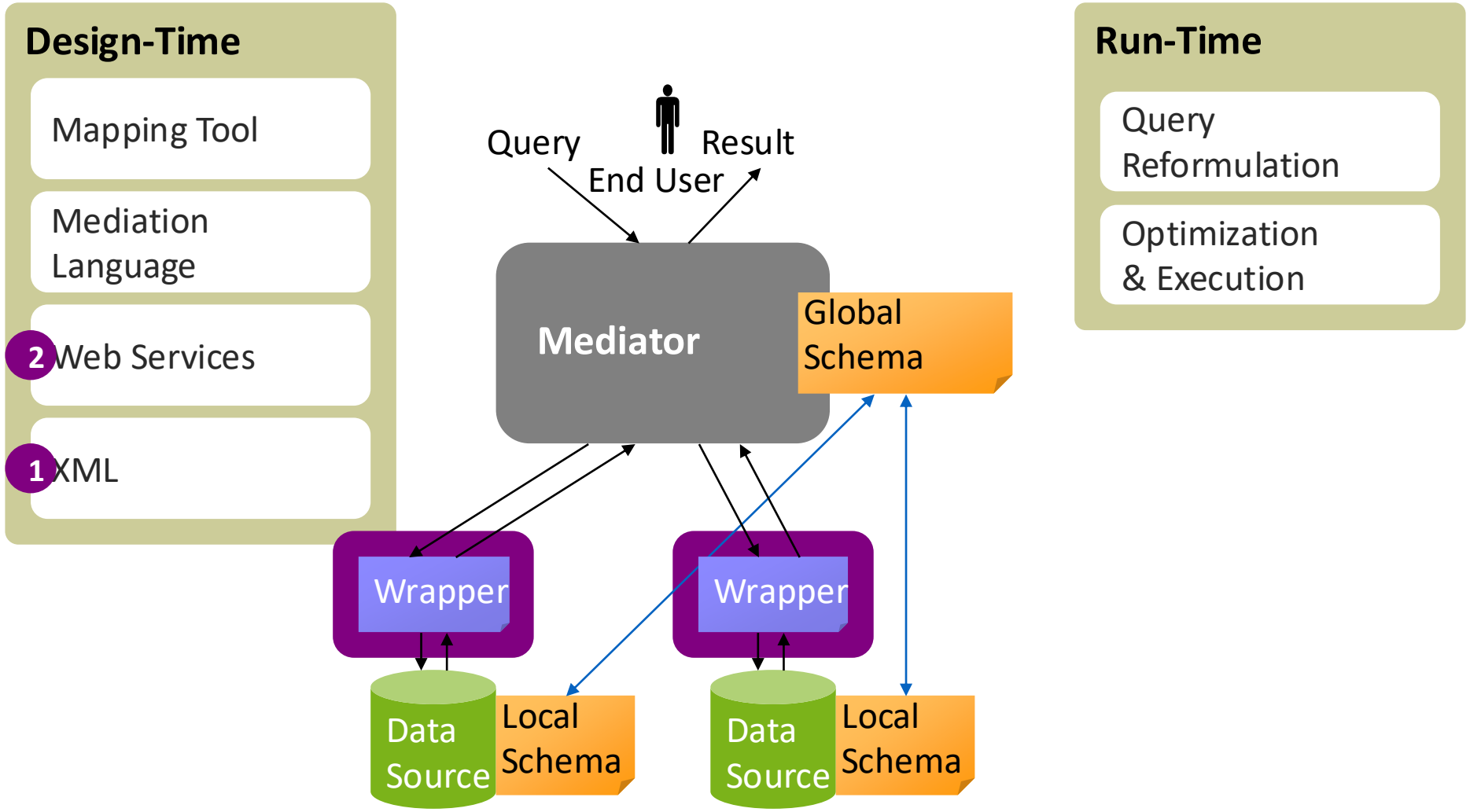
Mappings

Virtual Integration Architecture

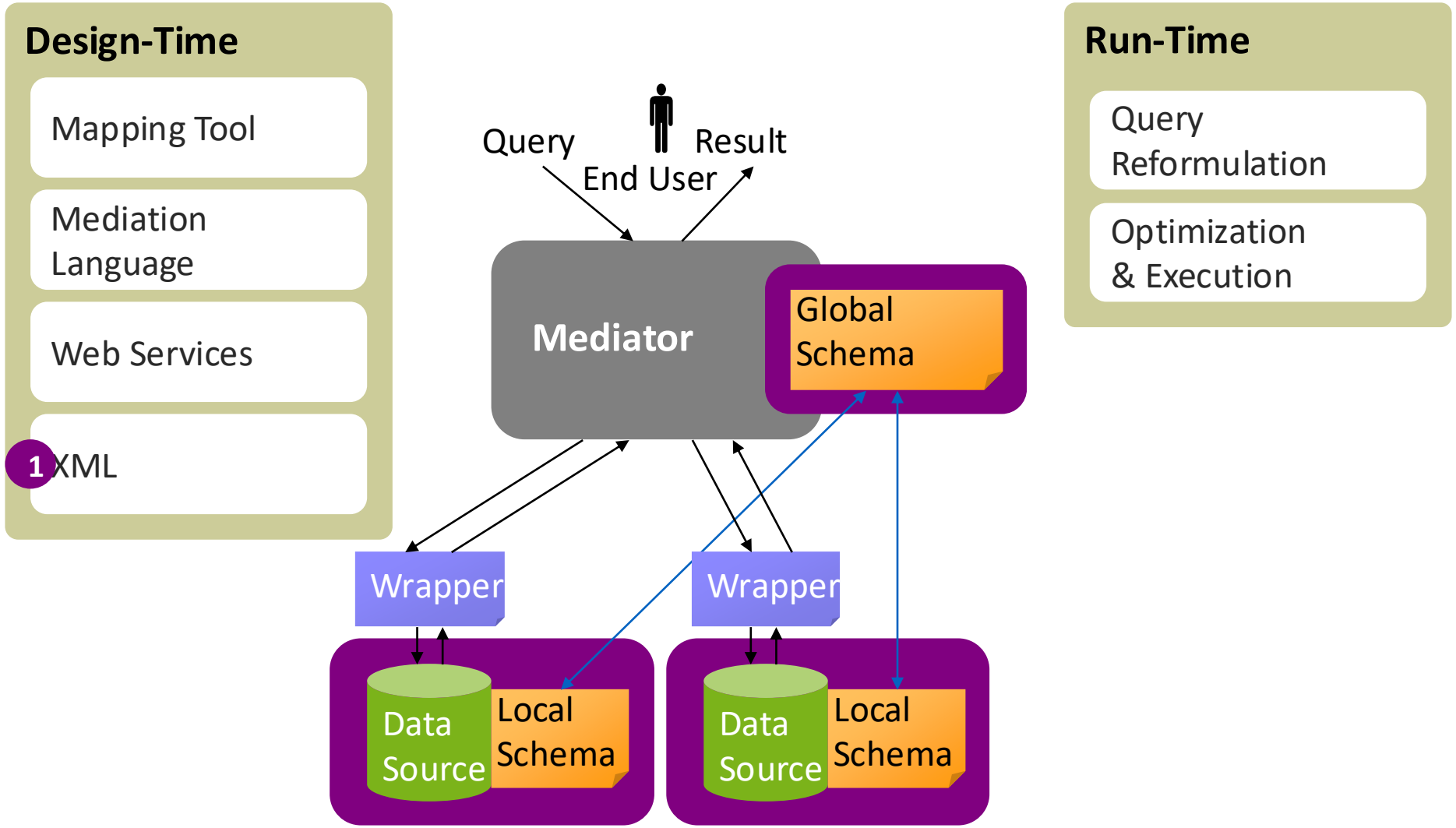
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On Demand Integration

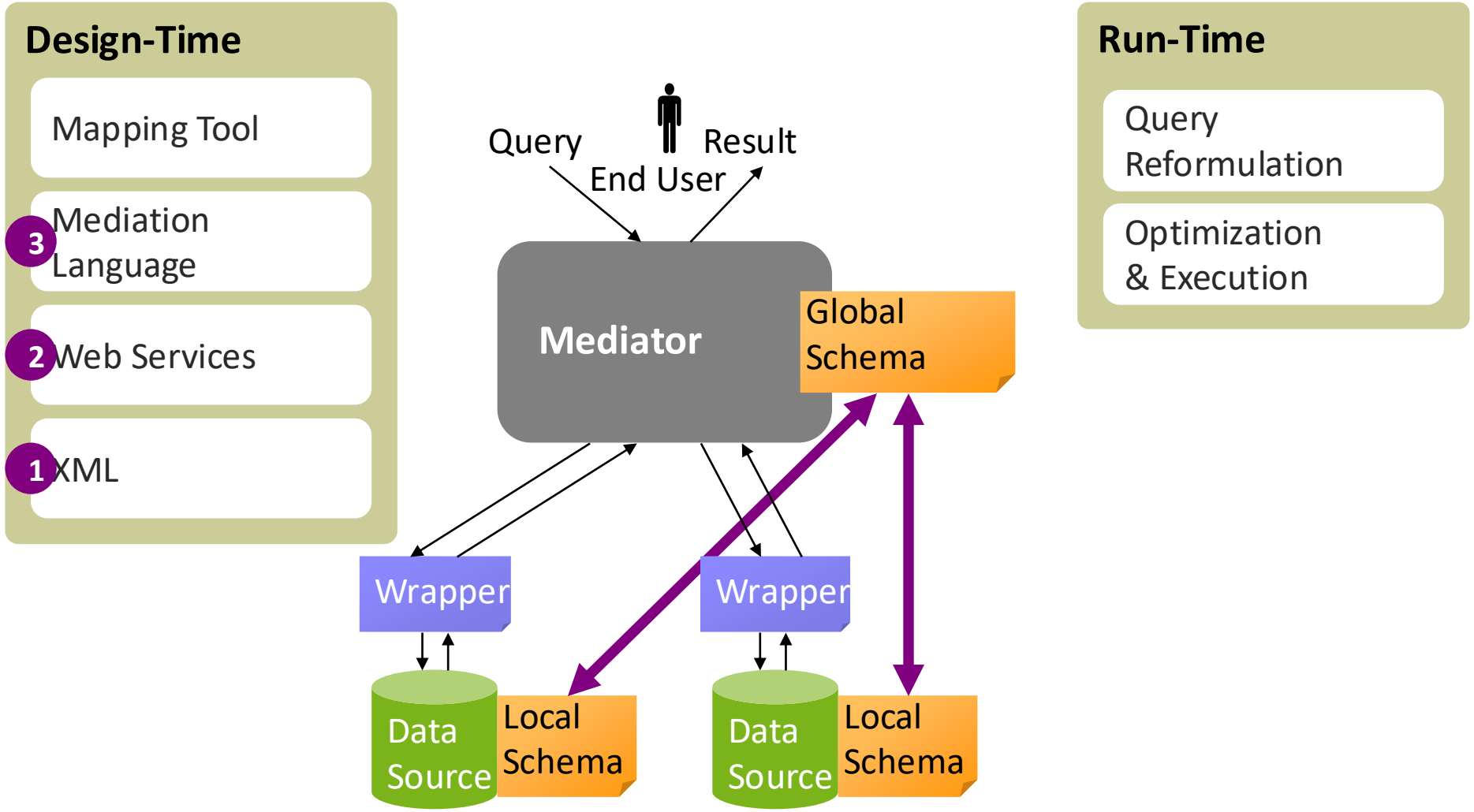
Virtual Integration Architecture



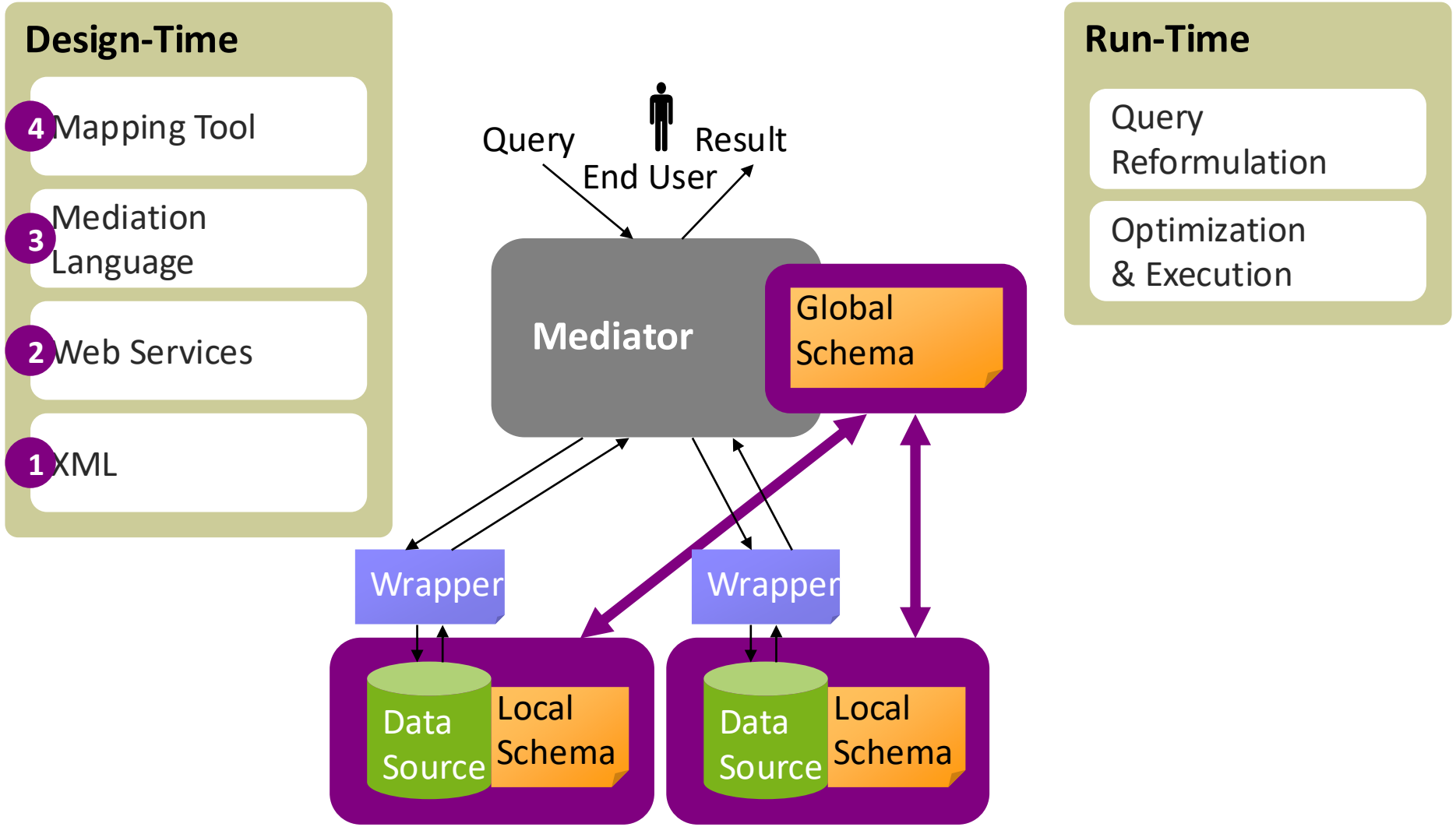
Virtual Integration Architecture



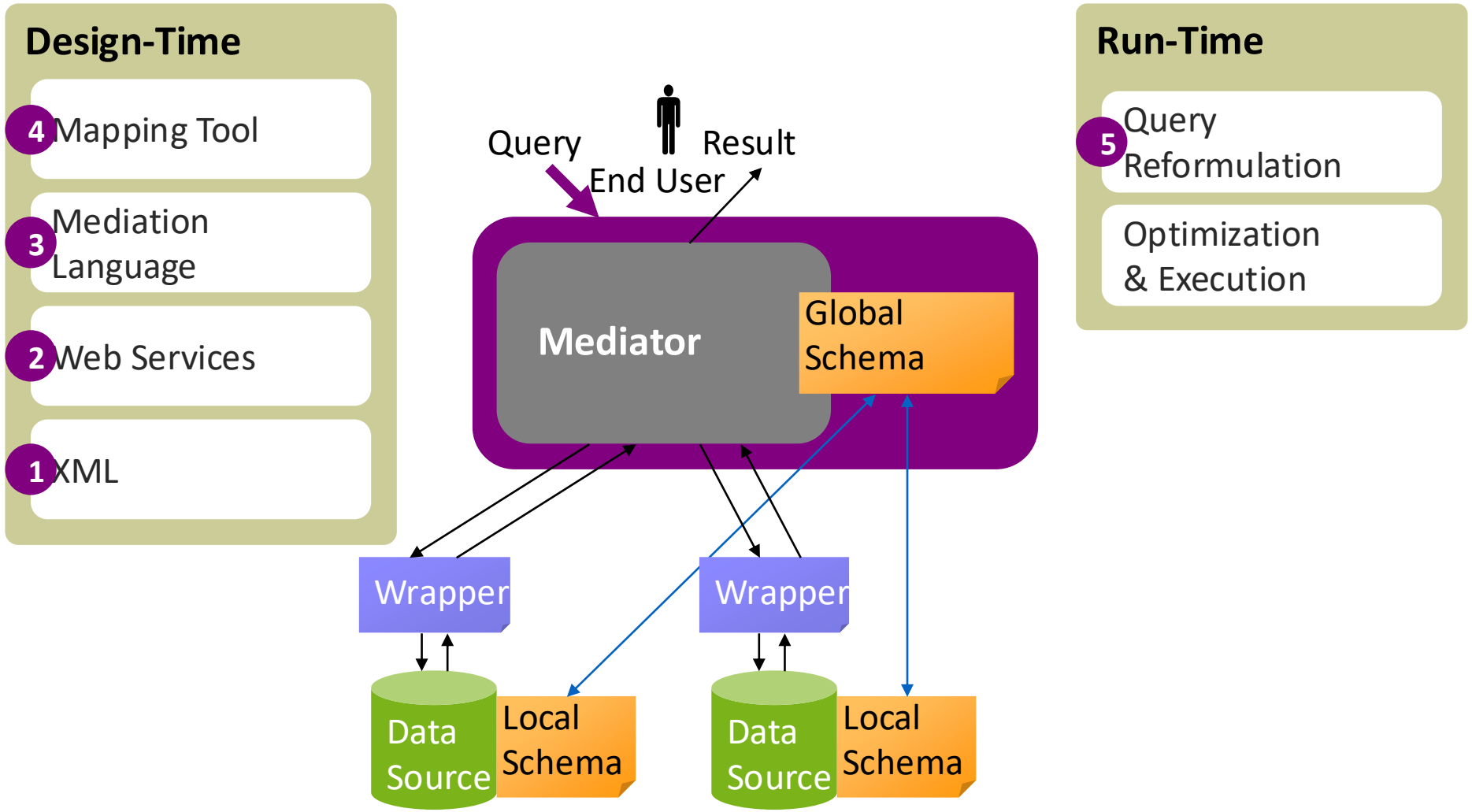
Virtual Integration Architecture



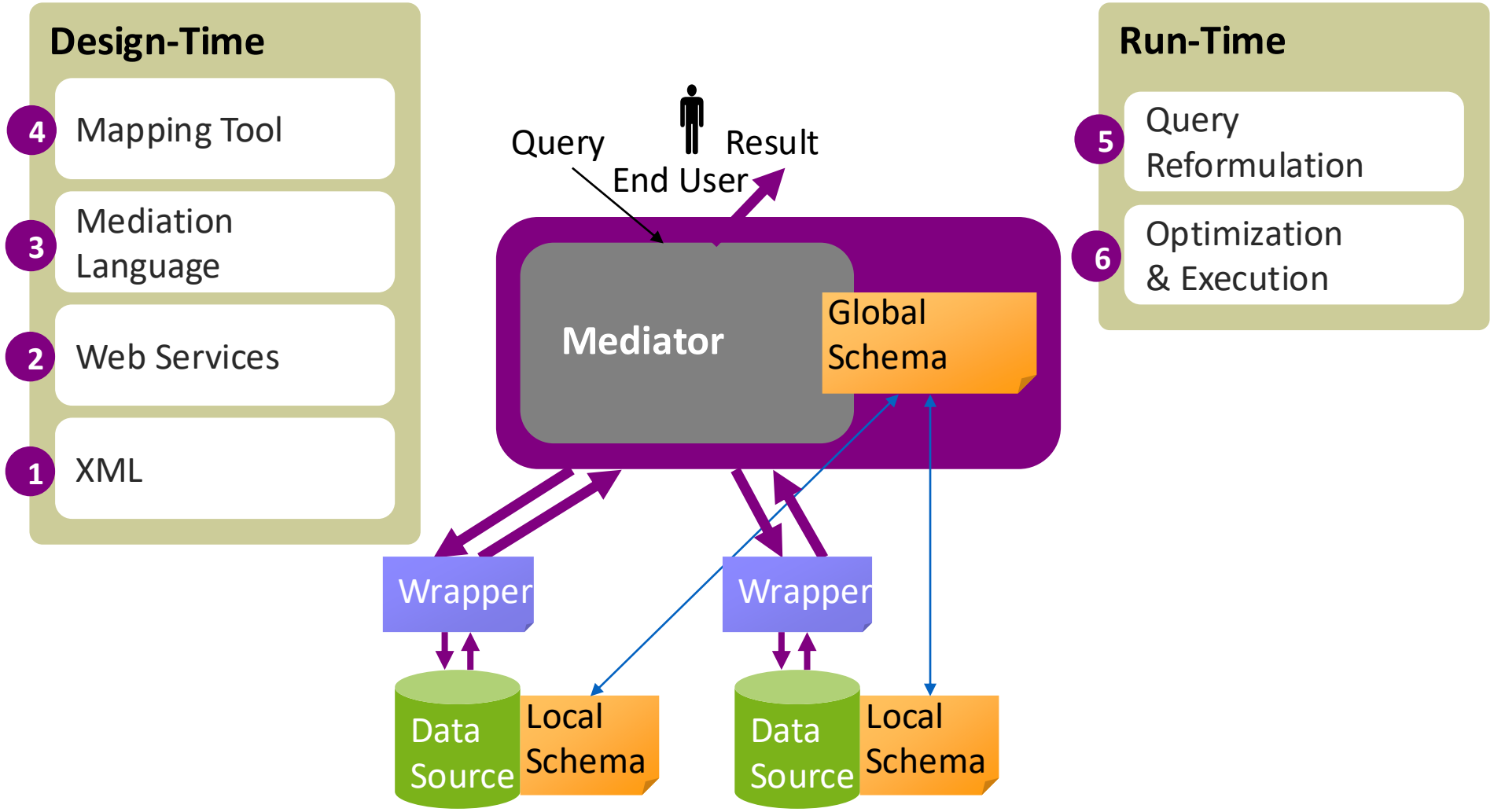
Virtual Integration Architecture



Virtual Integration Architecture



Virtual Integration Architecture



Virtual Integration Approaches

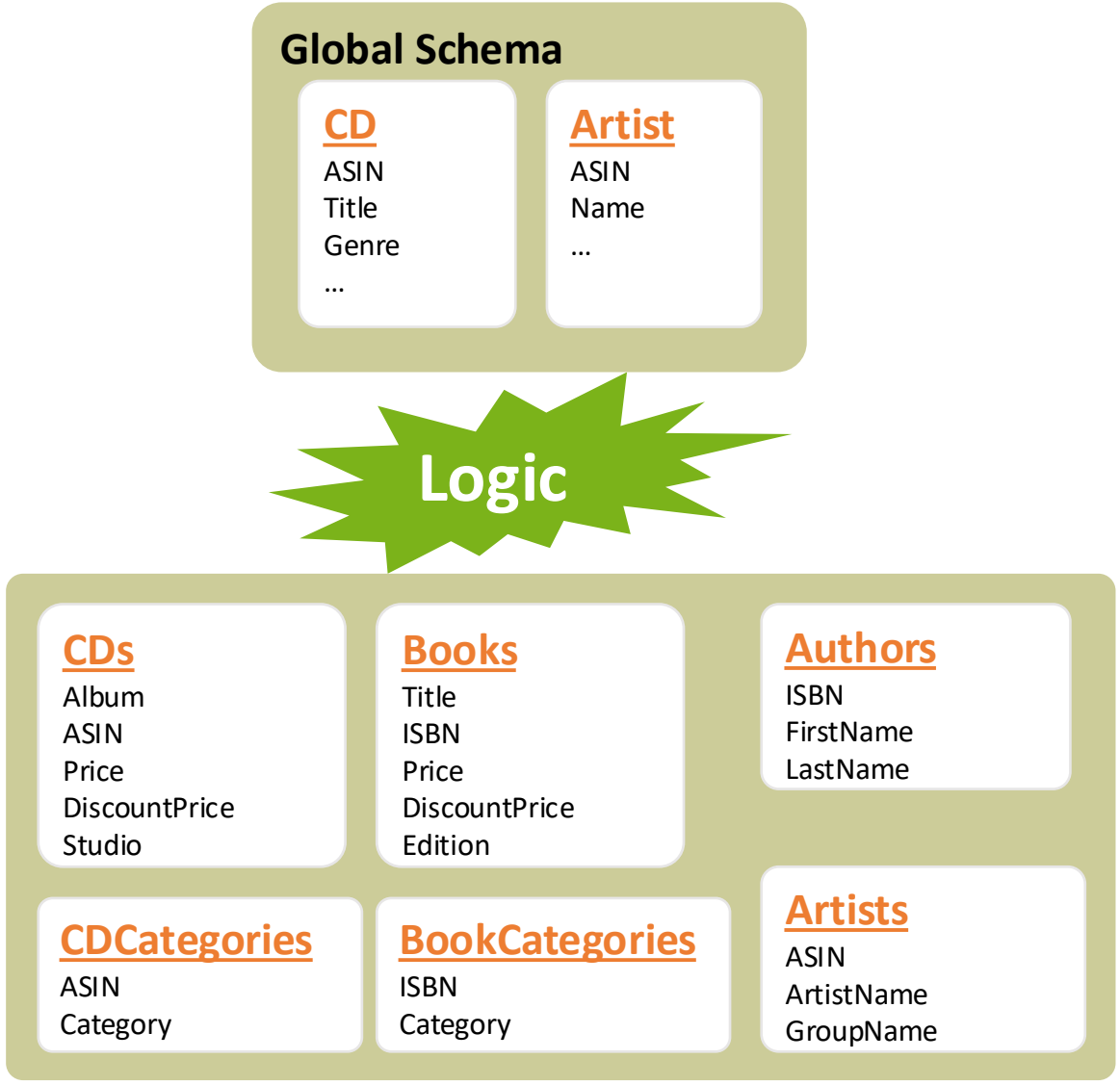
Dimensions to Consider:

- How many sources are we accessing?
- How autonomous are they?
- Meta-data about sources?
- Is the data structured?
- Queries or also updates?
- Requirements: accuracy, completeness, performance, handling inconsistencies.
- Closed world assumption vs. open world?

Requirements

- Relations and Attribute names
  - Names
- Tabular Organization
  - Splitting
- Granularity Level
  - More information
- Data Level Variations
  - Syntax
- Access Patterns Limitations
- Source completeness

Mediation Languages



Requirements from Source Descriptions

- Expressive power: distinguish between sources with closely related data. Hence, be able to prune access to irrelevant sources.
- Easy addition: make it easy to add new data sources.
- Reformulation: be able to reformulate a user query into a query on the sources efficiently and effectively.

Reformulation Problem

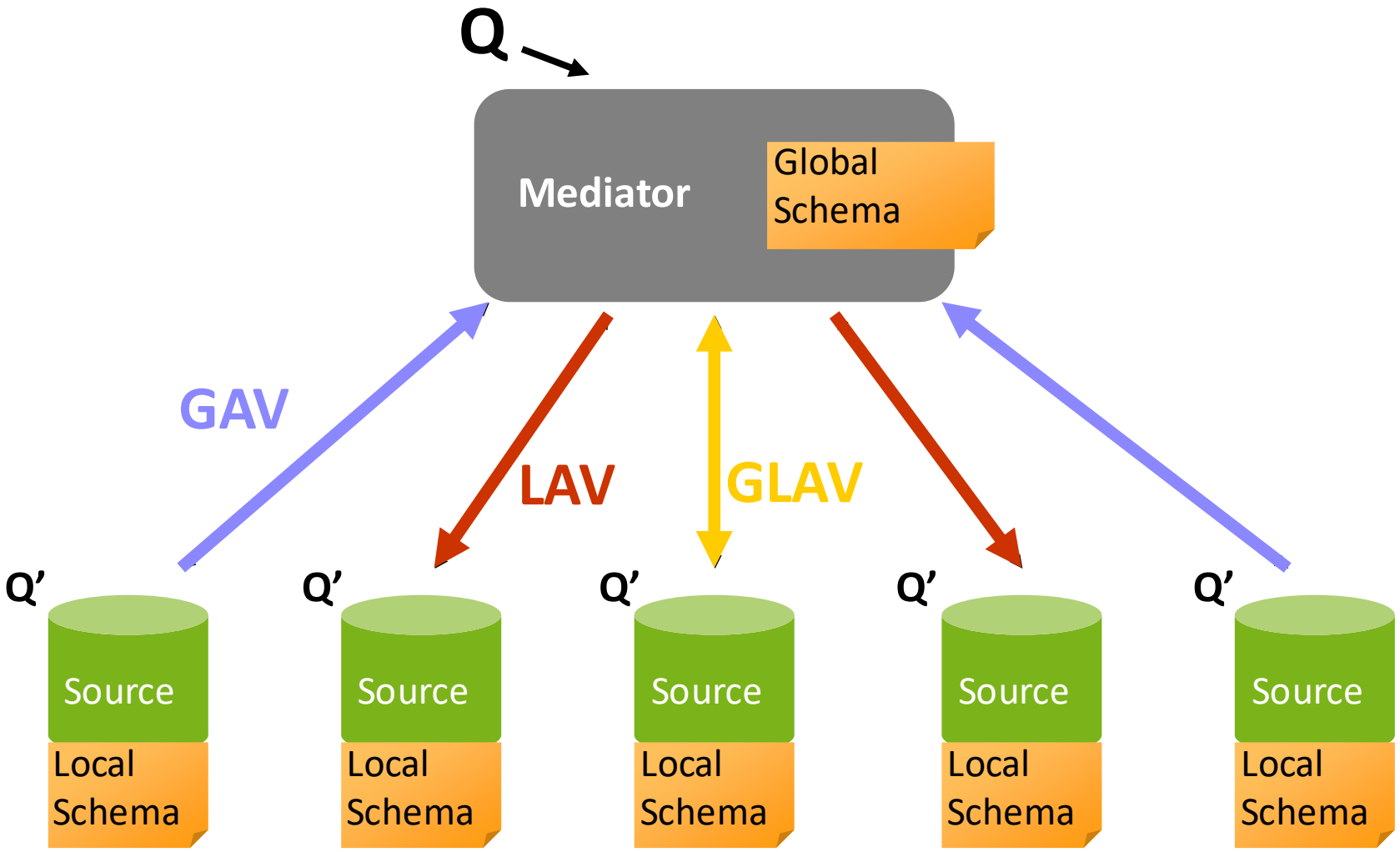
Given:

- A query Q posed over the global schema
- Descriptions of the data sources

Find:

- A query Q' over the data source relations, such that:
  - Q' provides only *correct answers* to Q, and
  - Q' provides *all* possible answers to Q given the sources.

Languages for Schema Mapping



Global-as-View (GAV)

Global Schema:

Movie(title, dir, year, genre)

Schedule(cinema, title, time)

Integrating View:

```
Create View Movie AS
    SELECT * FROM S1
    union
    SELECT * FROM S2
    union
    SELECT S3.title, S3.dir, S4.year, S4.genre
    FROM S3, S4
    WHERE S3.title = S4.title
```

[S1(title,dir,year,genre)]



[S2(title,dir,year,genre)]



[S3(title,dir)]



[S4(title,year,genre)]



Global-as-View: Example 2

Global Schema:

Movie(title, dir, year, genre)

Schedule(cinema, title, time)

Integrating View:

```
Create View Movie(title, dir, year, genre) AS
    SELECT title, dir, year, NULL
    FROM S1
    union
    SELECT title, dir, NULL, genre
    FROM S2
```

[S1(title,dir,year)]



[S2(title,dir,genre)]



Global-as-View Summary

- Query reformulation boils down to view unfolding.
- Very easy conceptually.
- Can build hierarchies of global schemas.
- You sometimes lose information. Not always natural.
- Adding sources is hard. Need to consider all other sources that are available.

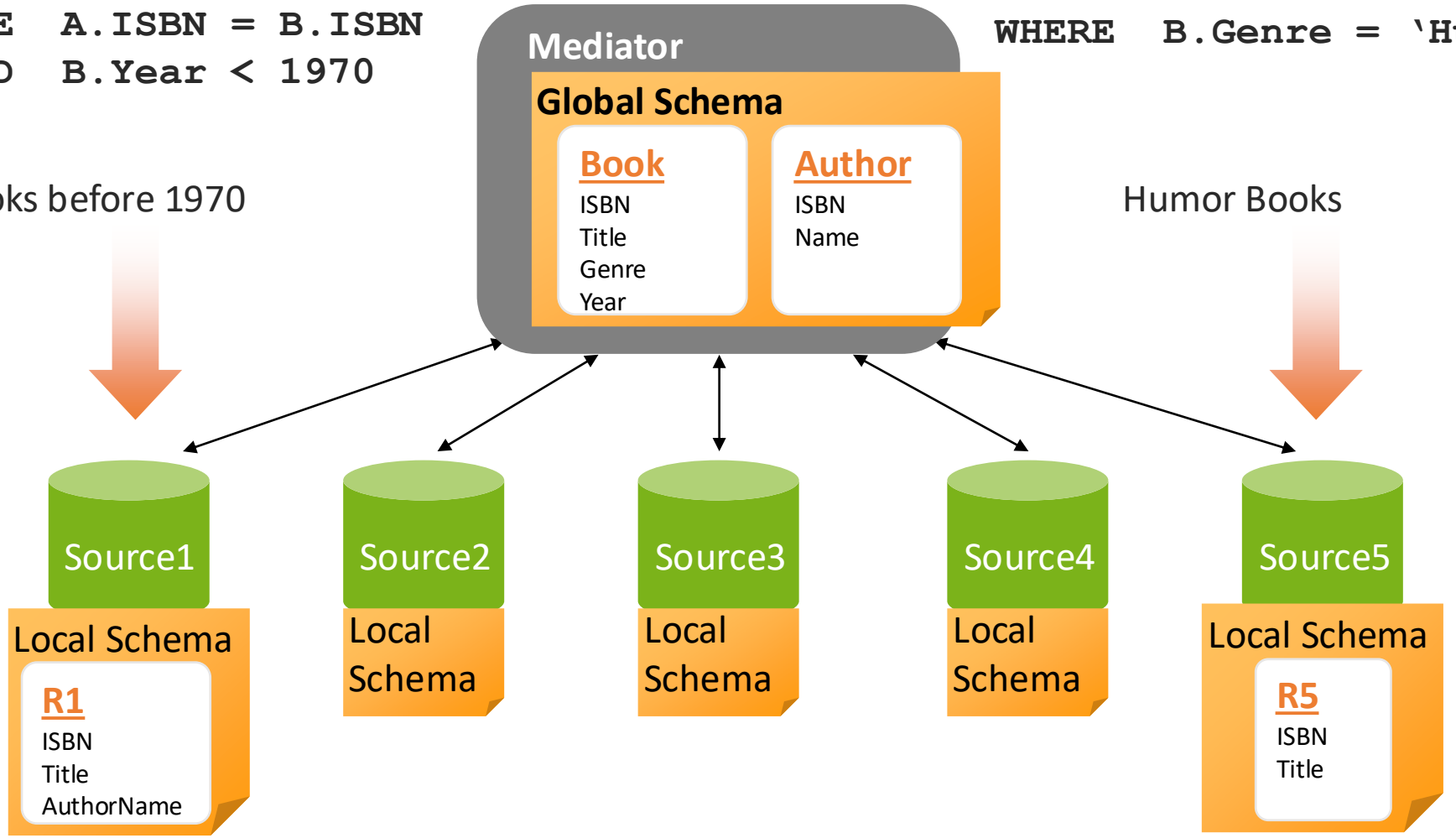
Local-as-View

- Sources are defined in terms of the global schema using expression
- Every source provides expressions on how it can generate pieces of the global schema
- Mediator can combine these expressions to find all possible ways to answer a query

Local-as-View (LAV)

```
Create View R1 AS
SELECT B.ISBN, B.Title, A.Name
FROM   Book B, Author A
WHERE  A.ISBN = B.ISBN
  AND  B.Year < 1970
```

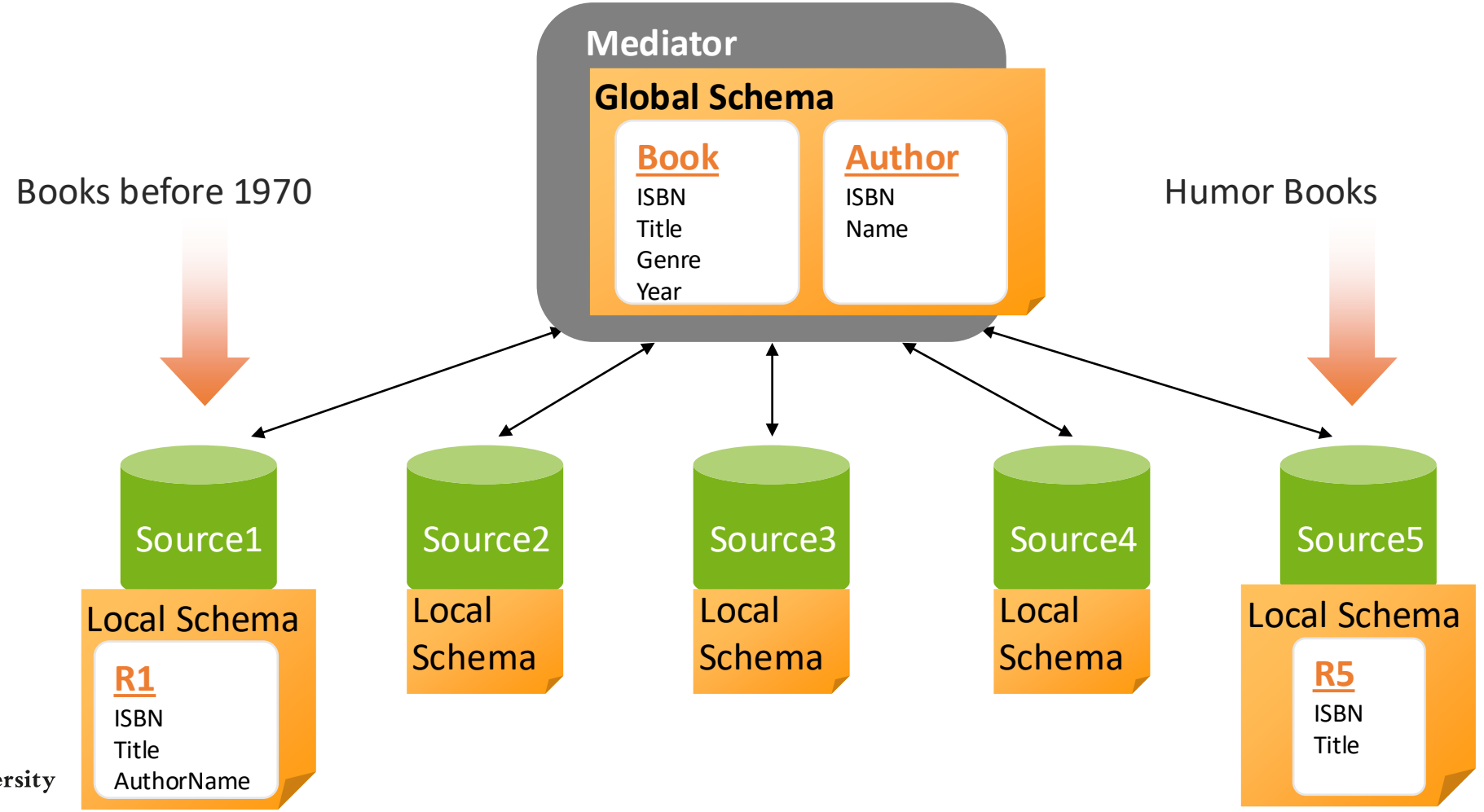
```
Create View R5 AS
SELECT B.ISBN, B.Title
FROM   Book B
WHERE  B.Genre = 'Humor'
```



Query Reformulation

Query: Find authors of humor books

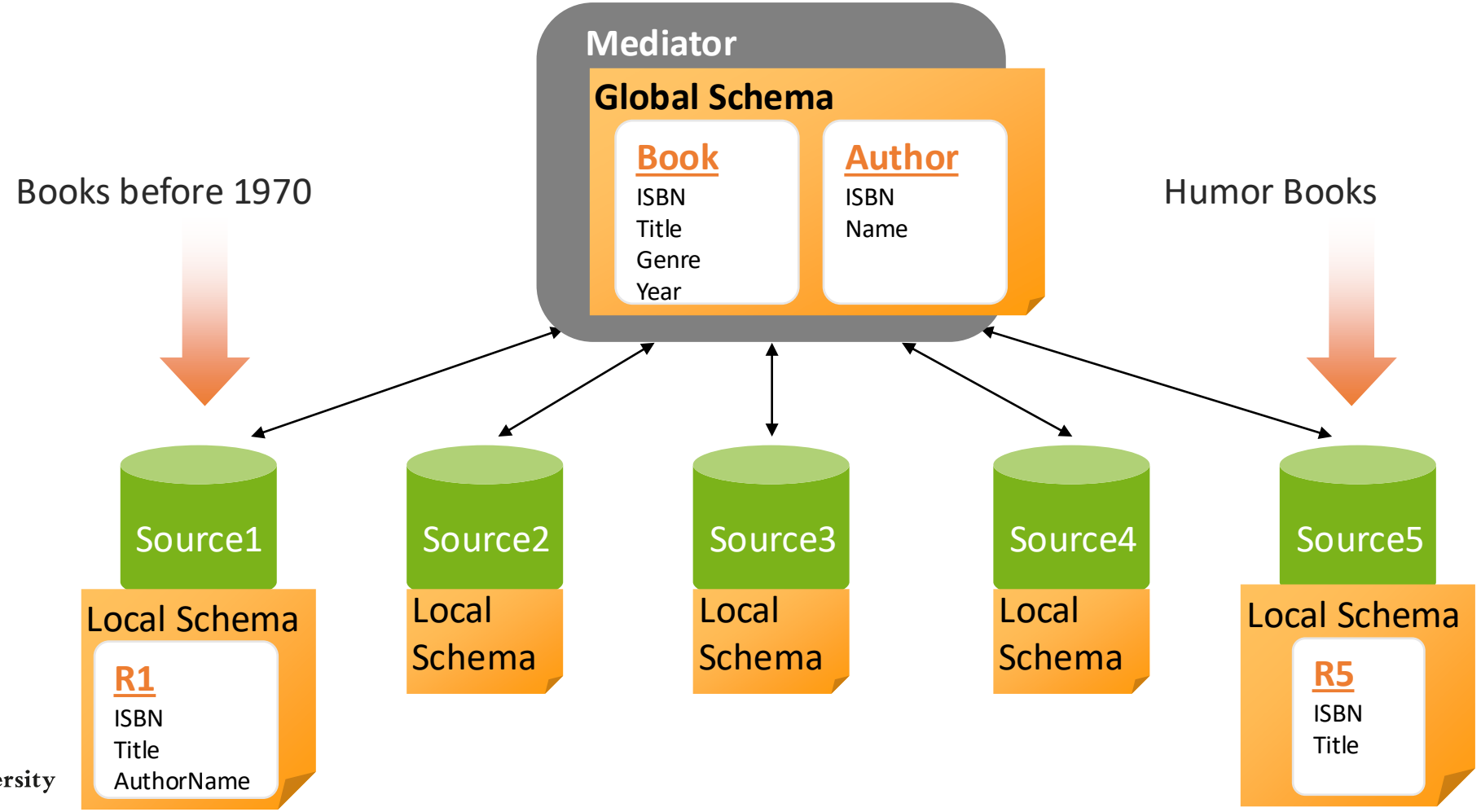
Plan: R1 Join R5



Query Reformulation

Query: Find authors of humor books before 1960

Plan: Can't do it!



Local-as-View: Example 1

Global Schema:

Movie(title, dir, year, genre)

Schedule(cinema, title, time)

Source Views:

```
Create Source S1 AS      [S1(title, dir, year, genre)]
    SELECT * FROM Movie
Create Source S3 AS      [S3(title, dir)]
    SELECT title, dir FROM Movie
Create Source S5 AS      [S5(title, dir, year)]
    SELECT title, dir, year
    FROM Movie
    WHERE year > 1960 AND genre='Comedy'
```

Local-as-View: Example 2

Global Schema:

Movie(title, dir, year, genre)

Schedule(cinema, title, time)

Source Views:

```
Create Source S4      [S4(cinema, genre)]
    SELECT cinema, genre
    FROM Movie M, Schedule S
    WHERE M.title=S.title
```

Now if we want to find
which cinemas are playing comedies, there is hope!

Local-as-View Summary

- Very flexible. You have the power of the entire query language to define the contents of the source.
- Hence, can easily distinguish between contents of closely related sources.
- Adding sources is easy: they're independent of each other.
- Query reformulation: *answering queries using views!*

Additional References

- [Lenz02] Maurizio Lenzerini, Data integration: a theoretical perspective, PODS2002, https://dl.acm.org/doi/10.1145/543613.543644.
- [Mill18] Renee J. Miller, Open Data Integration, VLDB2018, http://www.vldb.org/pvldb/vol11/p2130-miller.pdf



- Summarize what you learned today in 2-minutes

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